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Steele et al.

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(54) **HIGH-PRESSURE MULTILATERAL JUNCTION WITH MAINBORE AND LATERAL ACCESS AND CONTROL**

(58) **Field of Classification Search**
CPC ... E21B 17/18; E21B 41/0035; E21B 41/0042
See application file for complete search history.

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Related U.S. Application Data

(60) Provisional application No. 62/946,219, filed on Dec. 10, 2019.

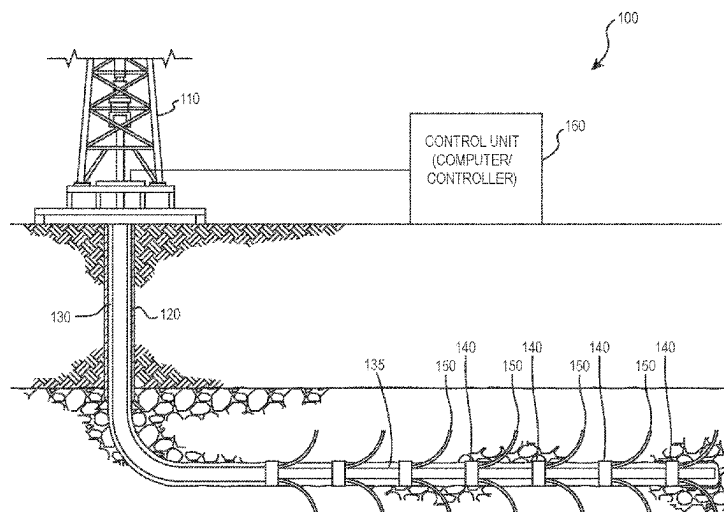
(51) **Int. Cl.**
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E21B 17/18 (2006.01)
(Continued)

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CPC **E21B 41/0035** (2013.01); **E21B 17/18**
(2013.01); **E21B 23/00** (2013.01);
(Continued)

(57) **ABSTRACT**

Provided is a y-block, a multilateral junction, and a method for forming a well system. The y-block, in one aspect, includes a housing having a first end and a second opposing end, and a single first bore extending into the housing from the first end, the single first bore defining a first centerline. The y-block according to this aspect further includes second and third separate bores extending into the housing and branching off from the single first bore, the second bore defining a second centerline and the third bore defining a third centerline, wherein the second and third centerlines are angled relative to one another.

36 Claims, 19 Drawing Sheets



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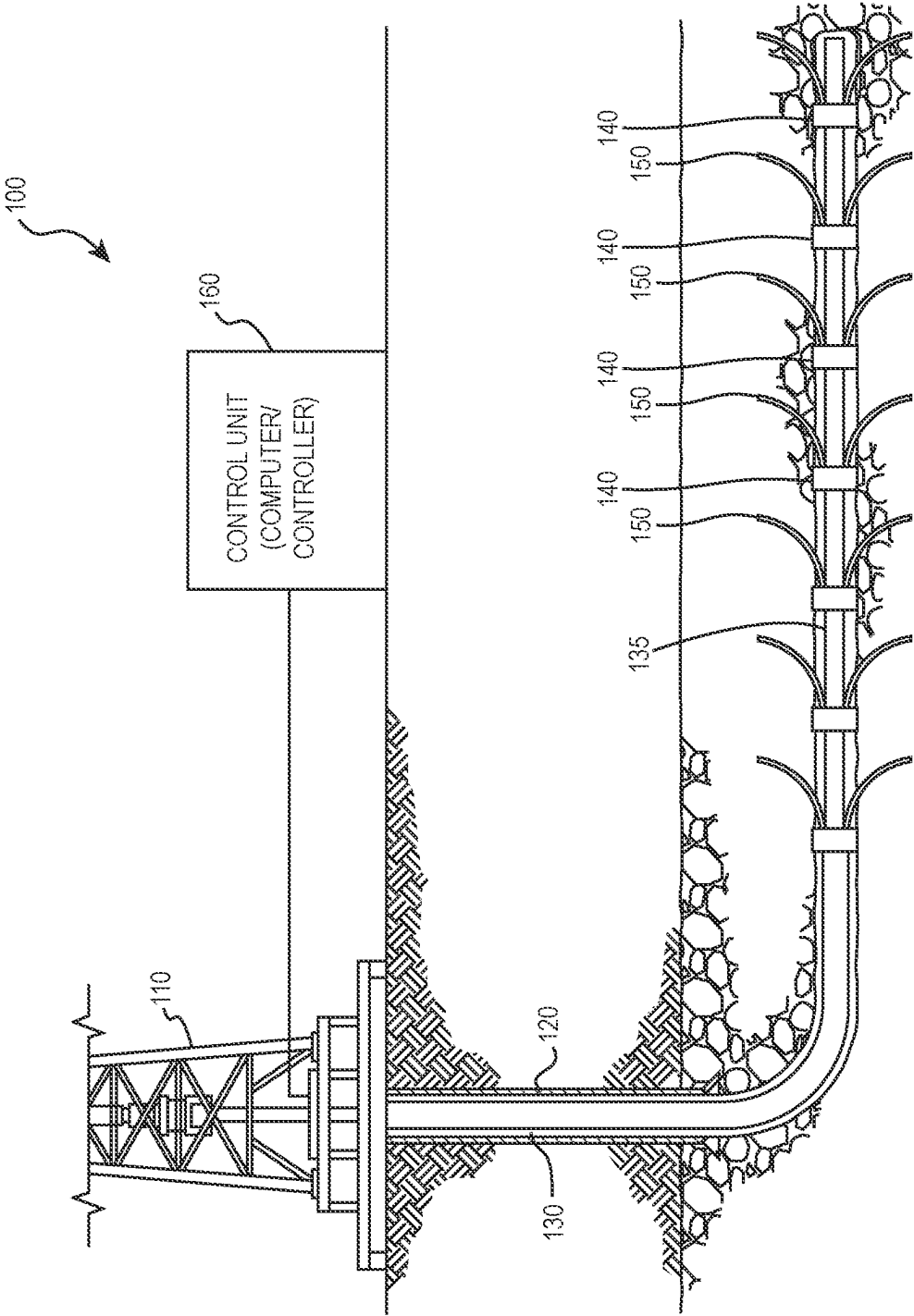


FIG. 1

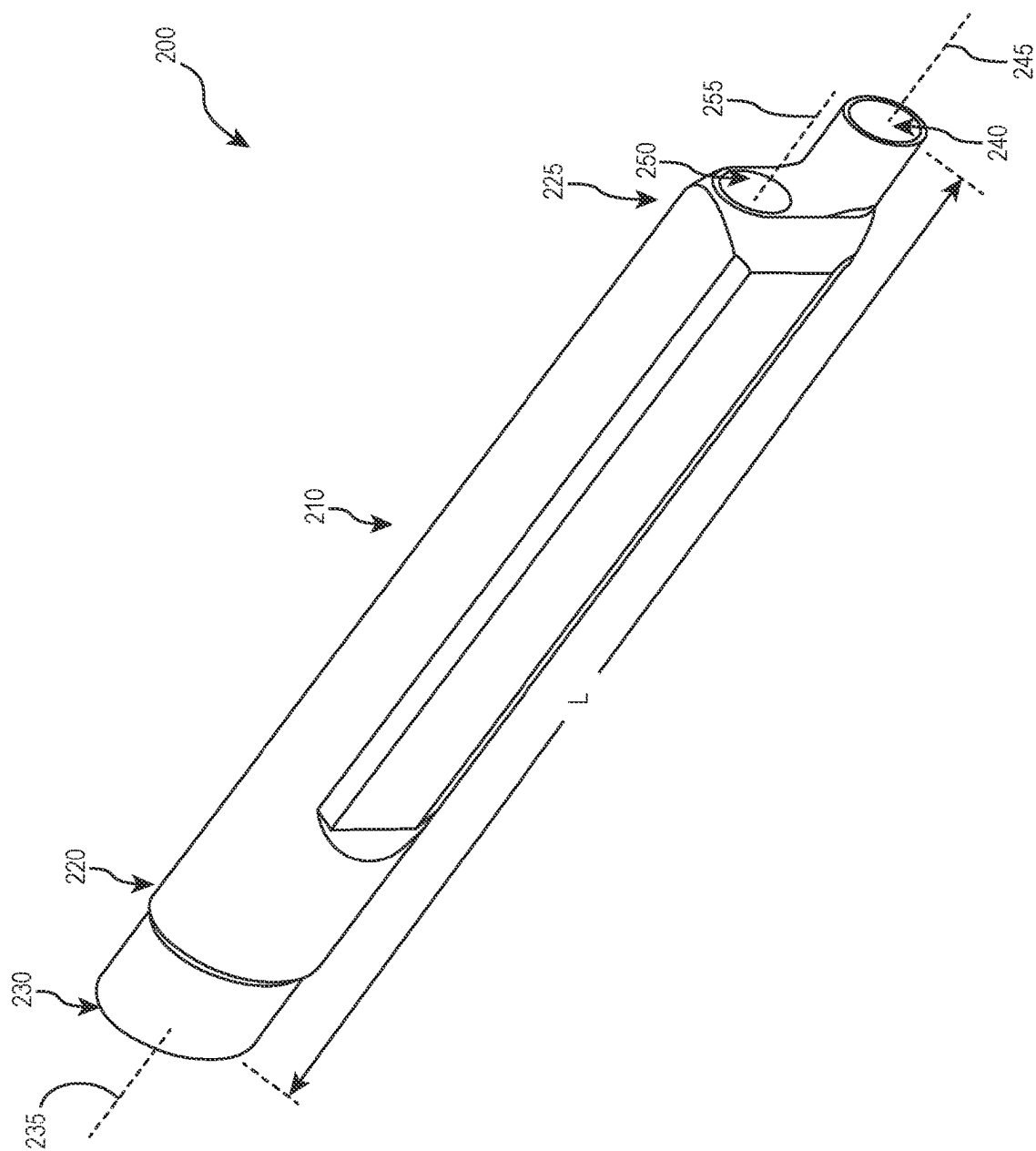
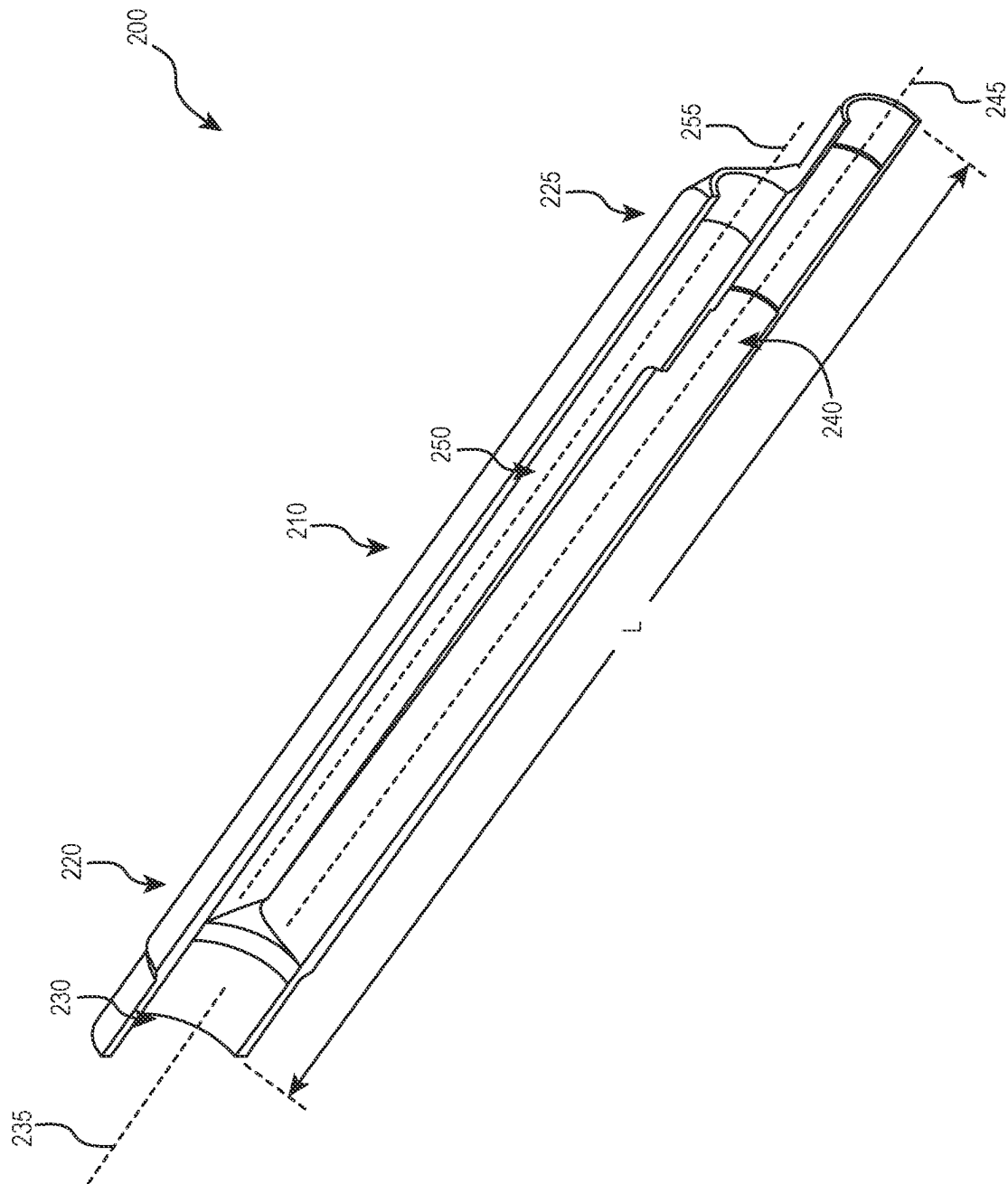


FIG. 2



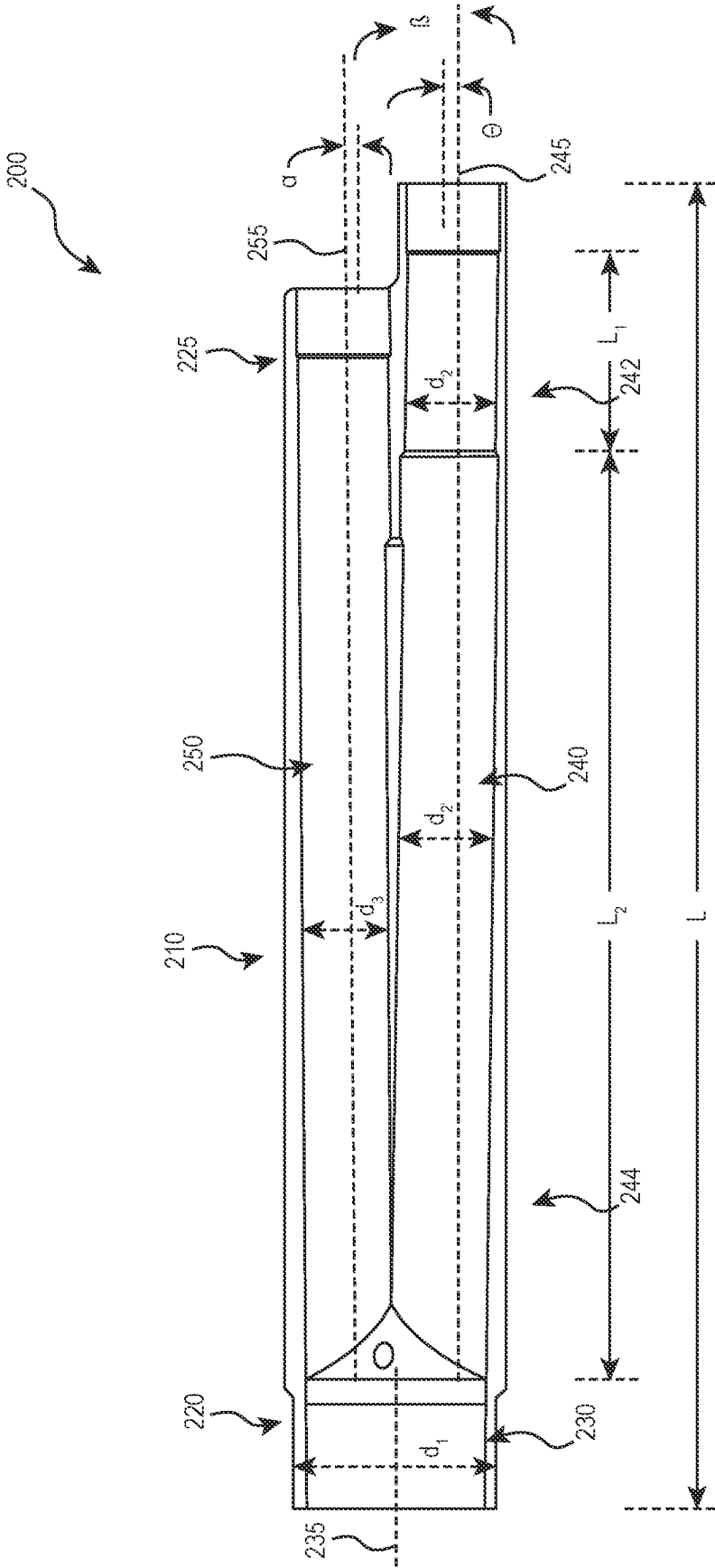


FIG. 4

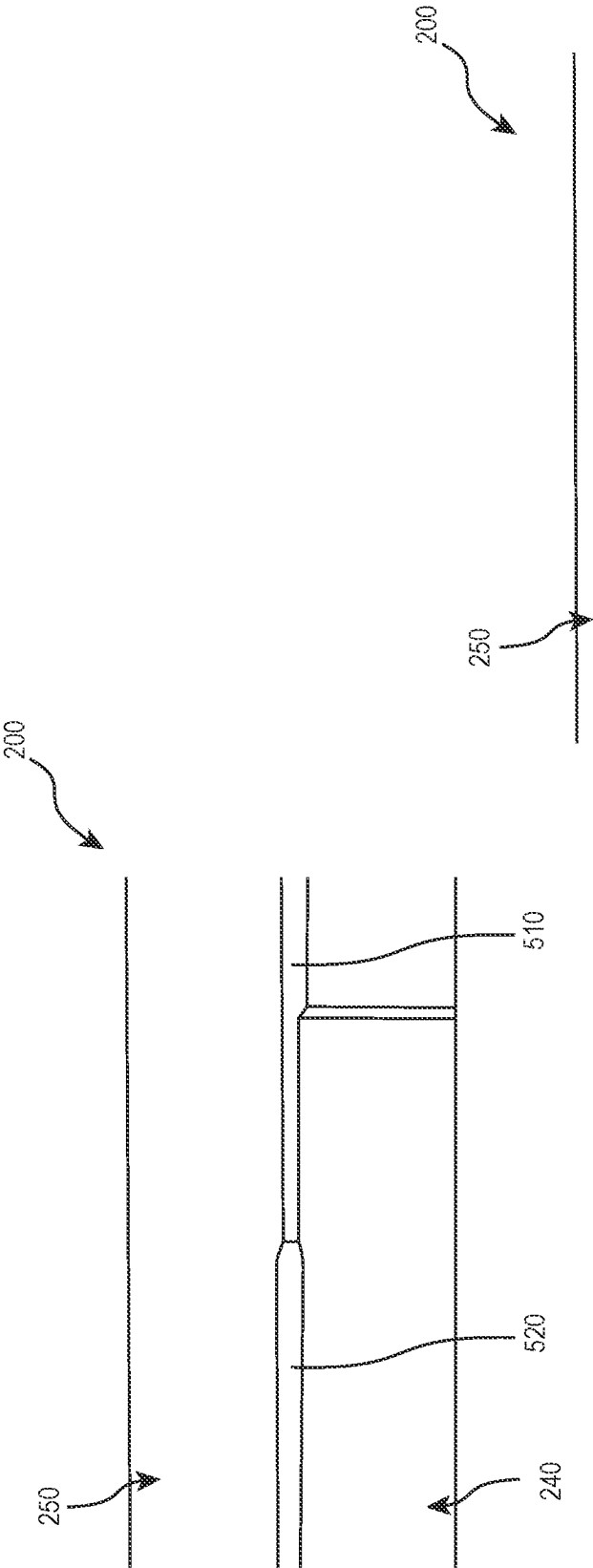


FIG. 5A

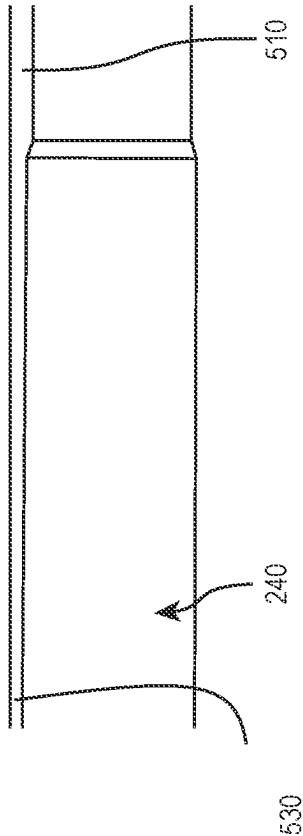


FIG. 5B

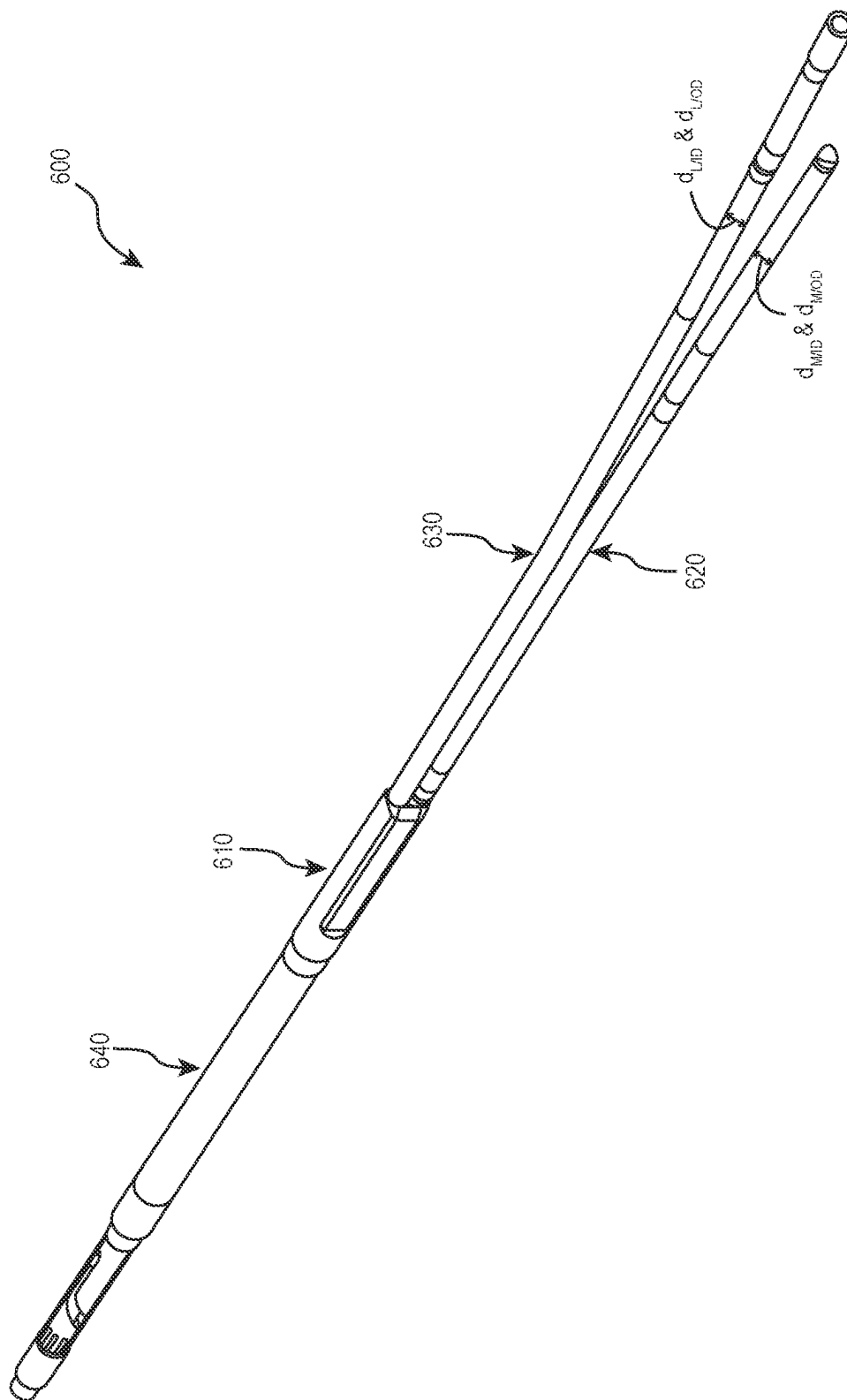


FIG. 6

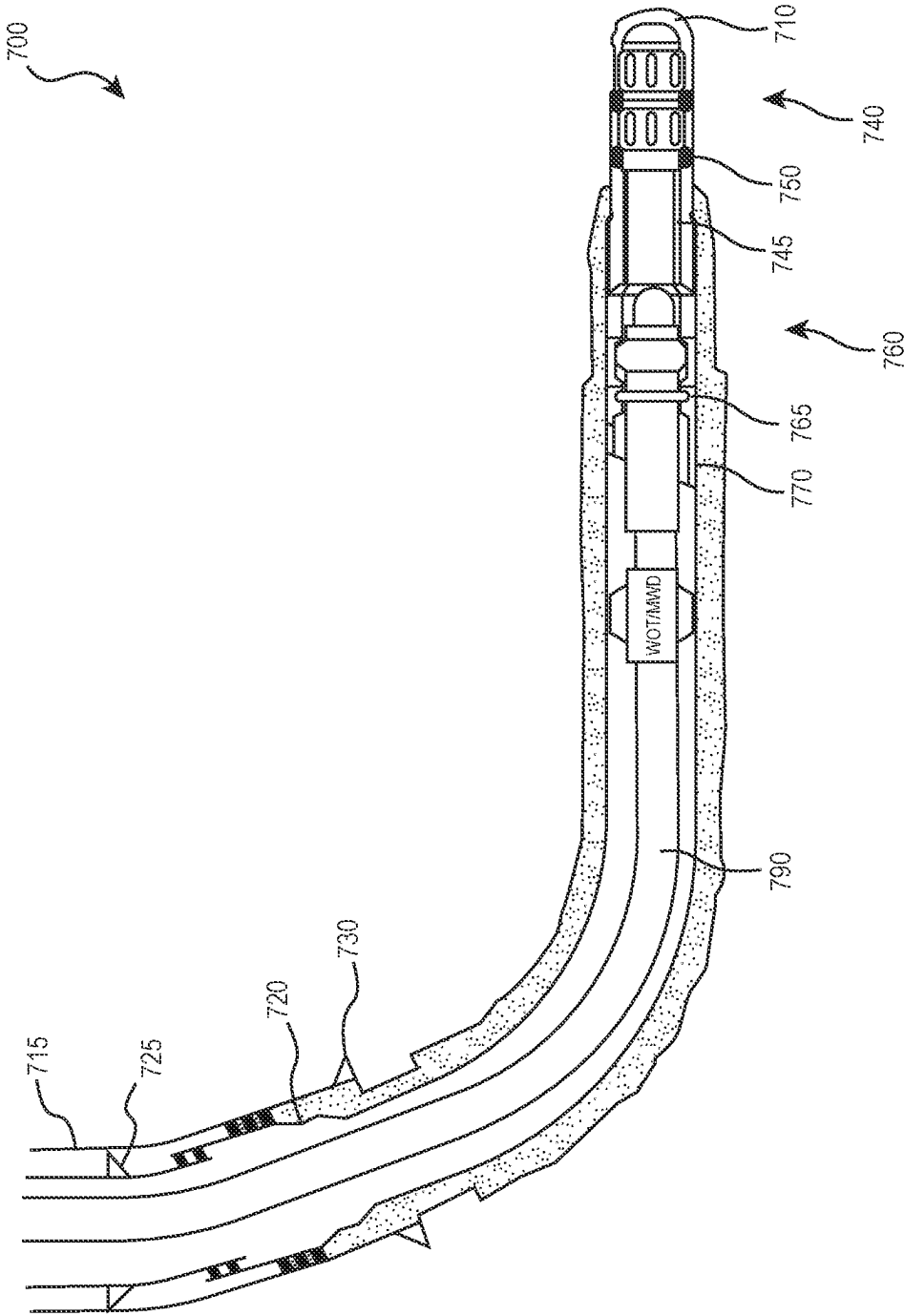


FIG. 7

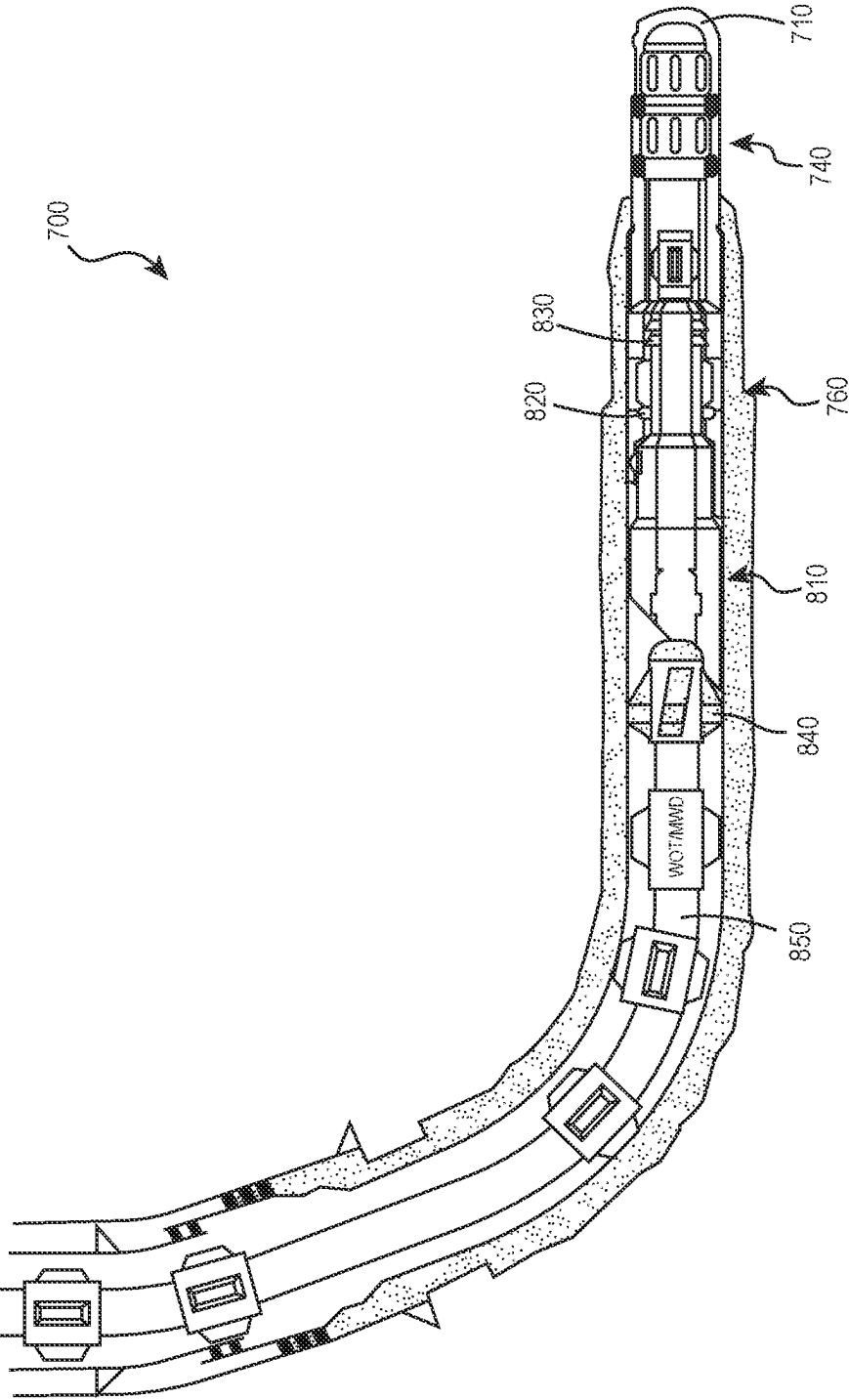


FIG. 8

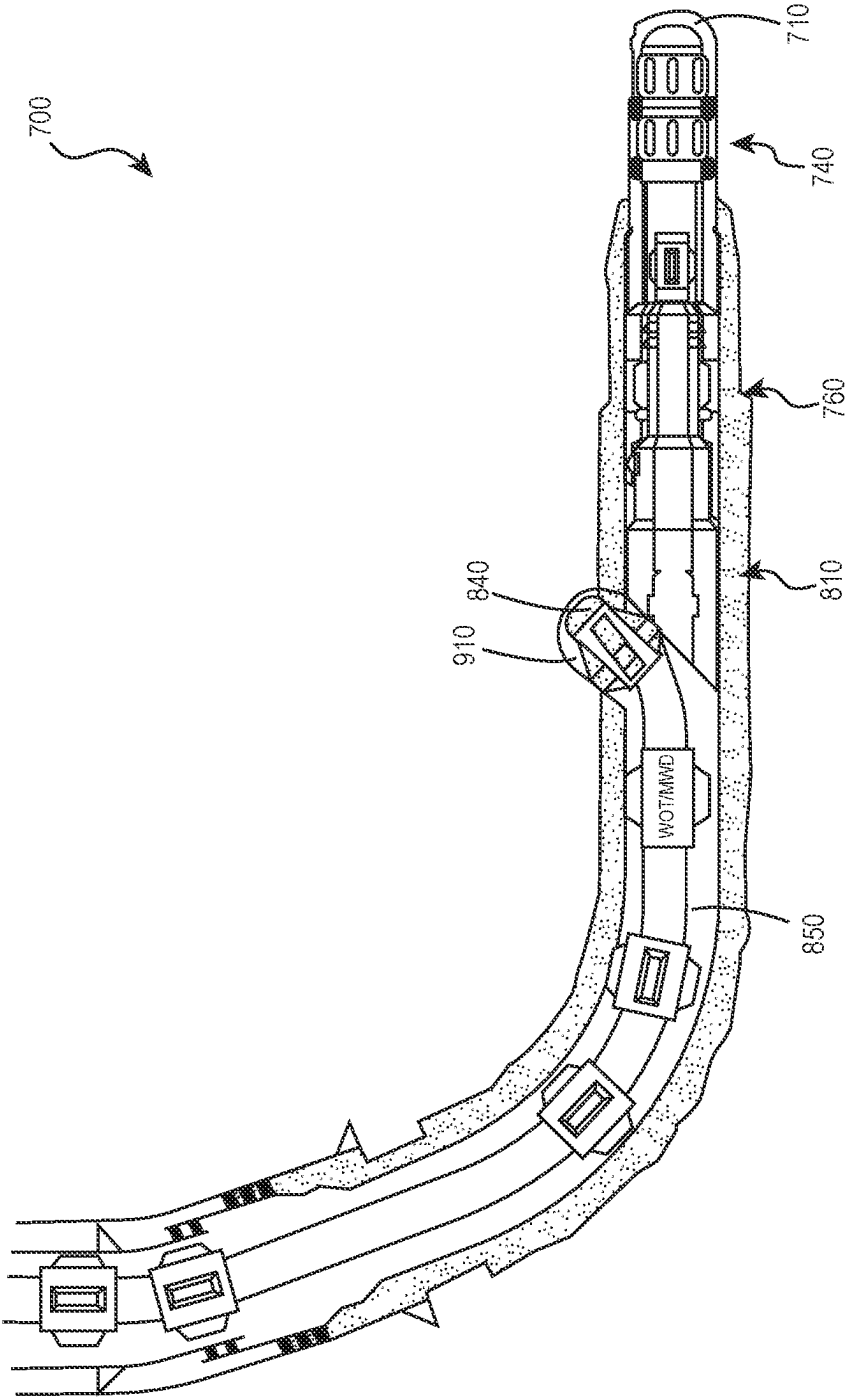


FIG. 9

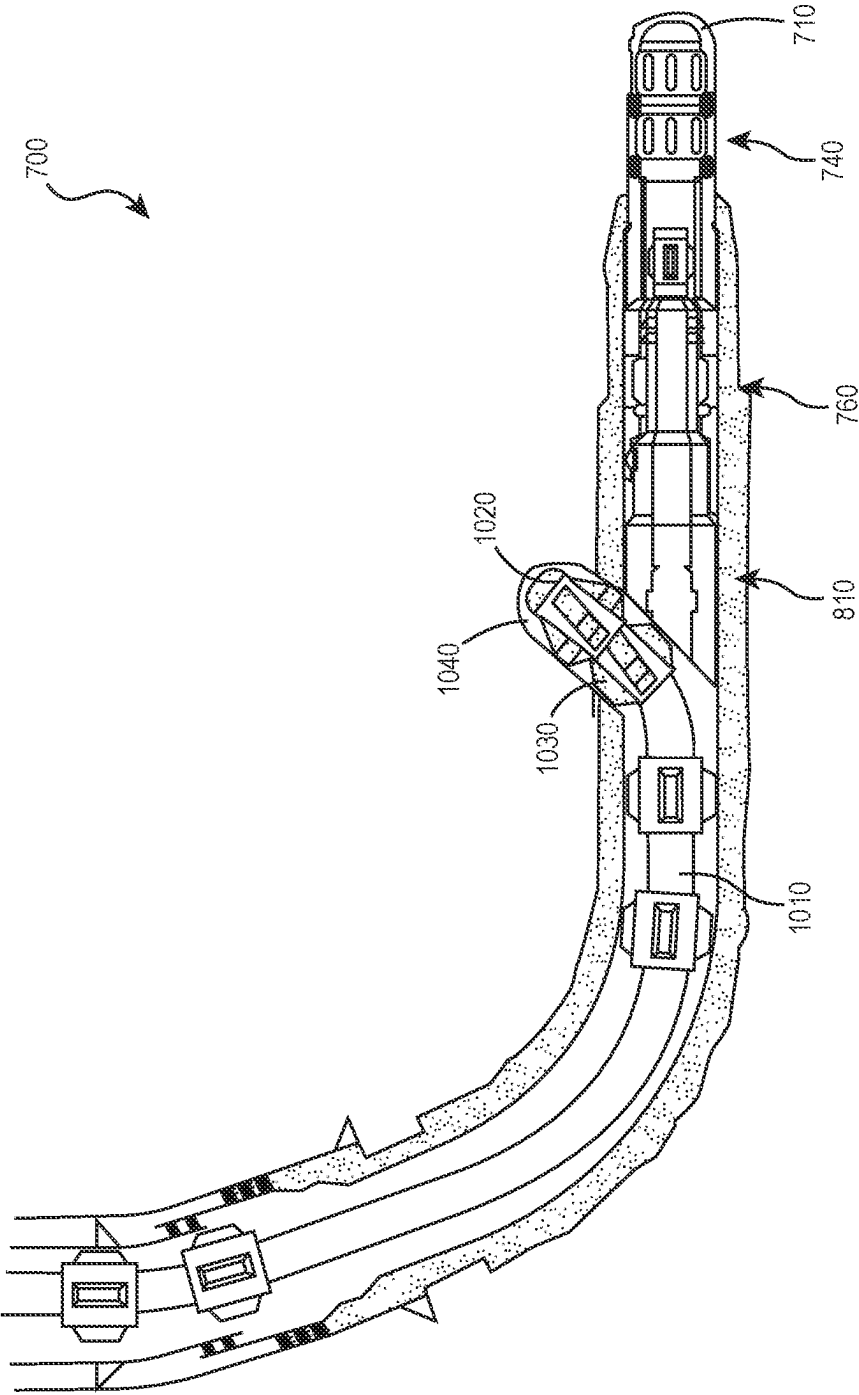


FIG. 10

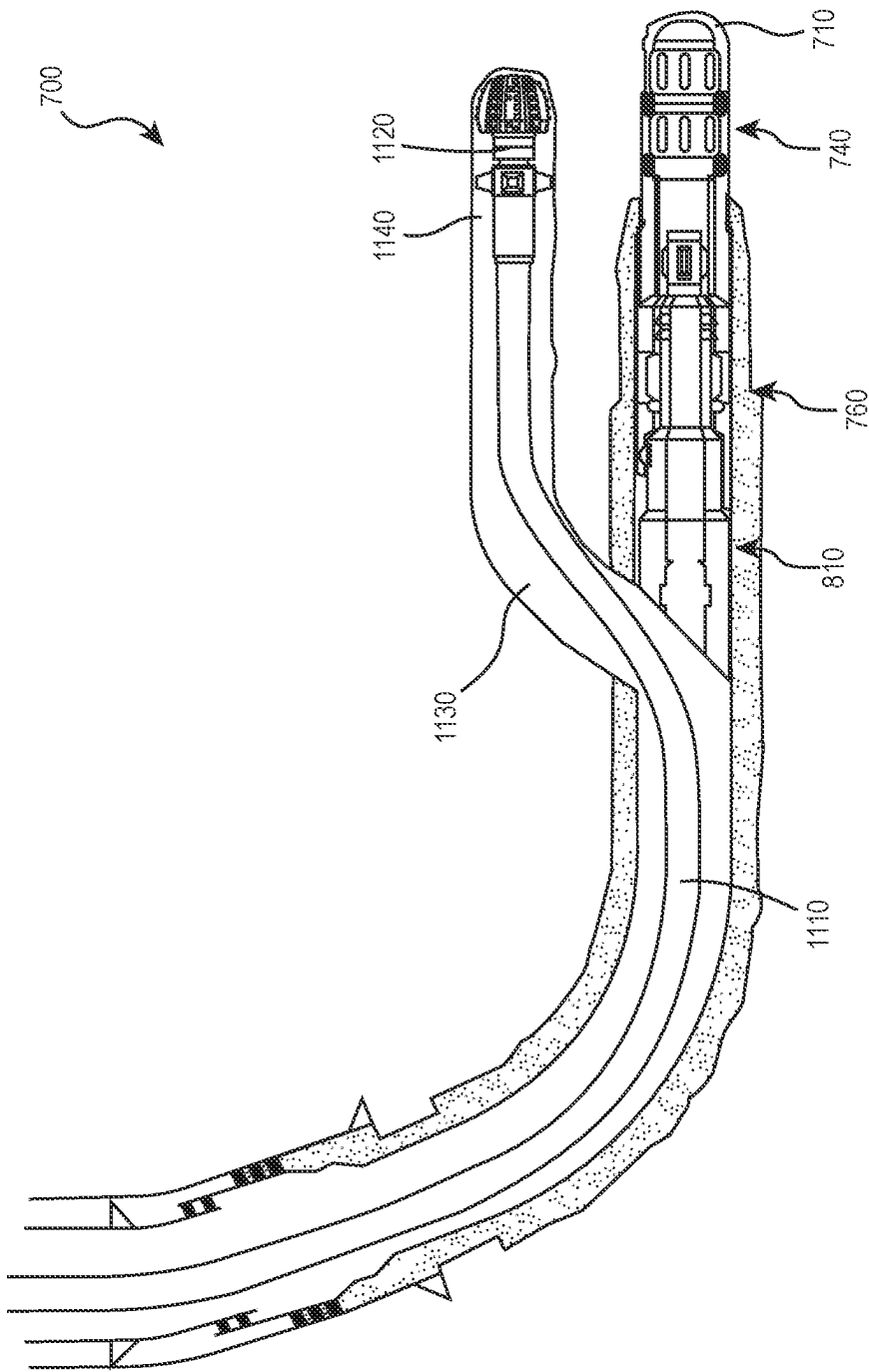
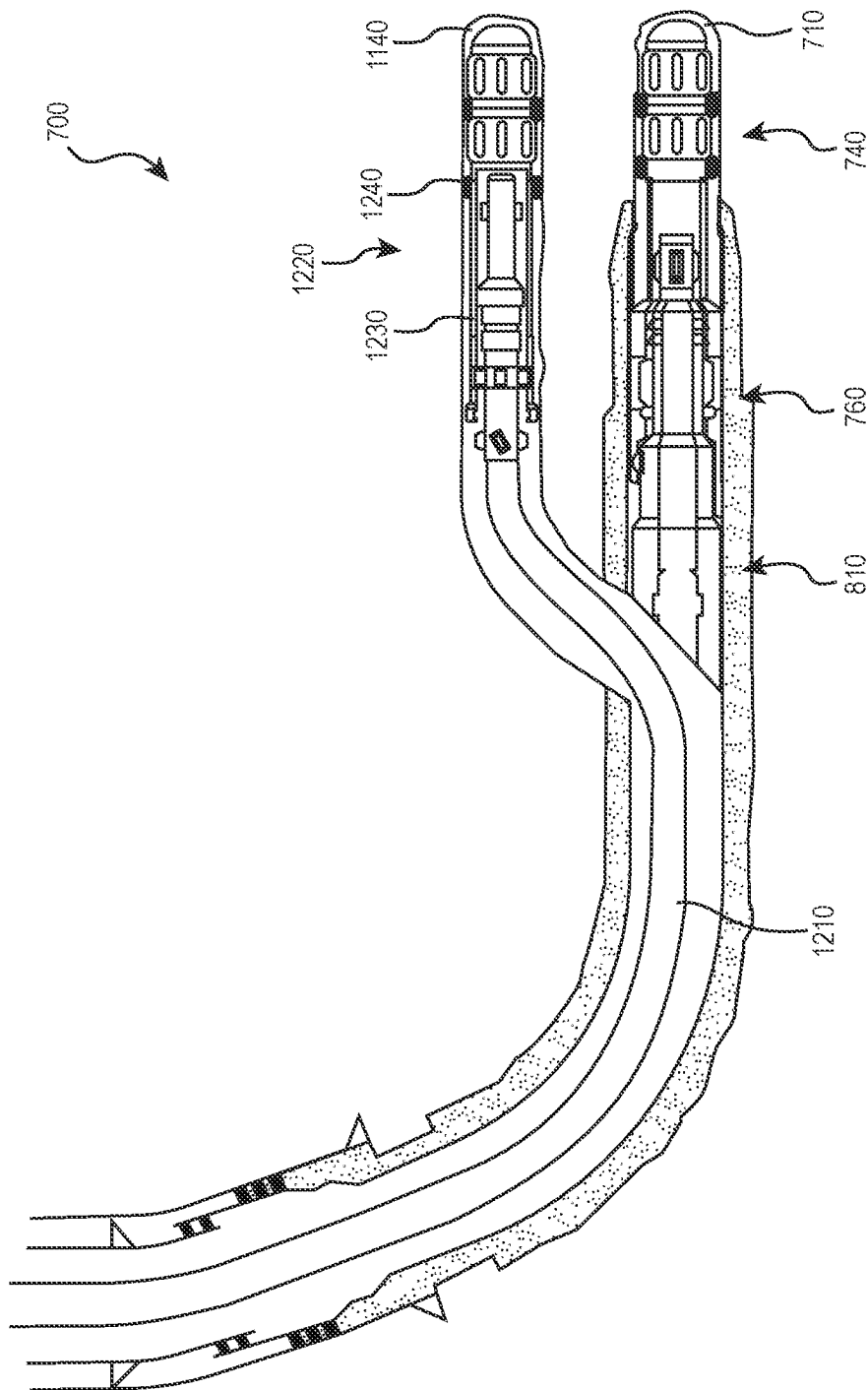
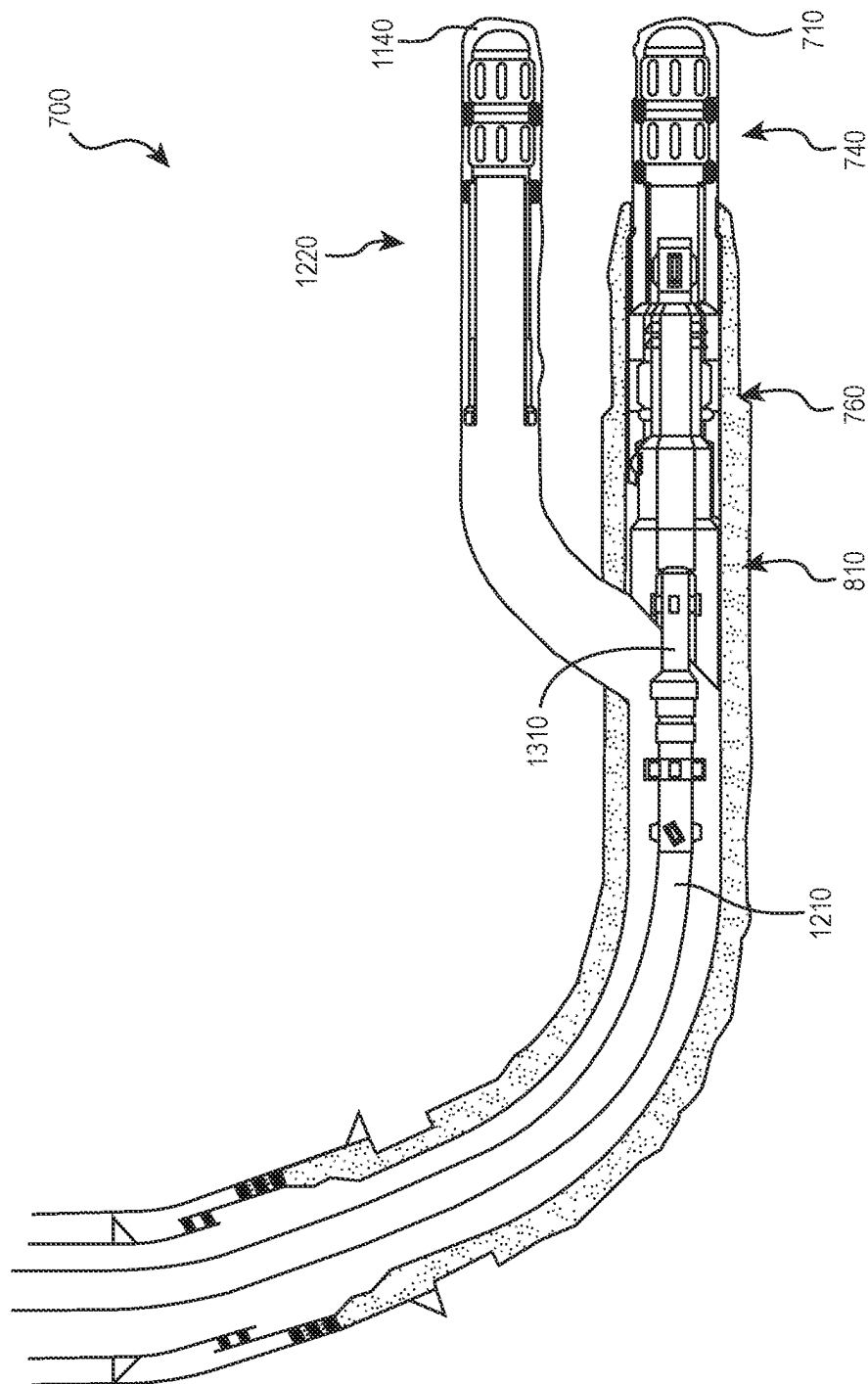


FIG. 11



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G
F



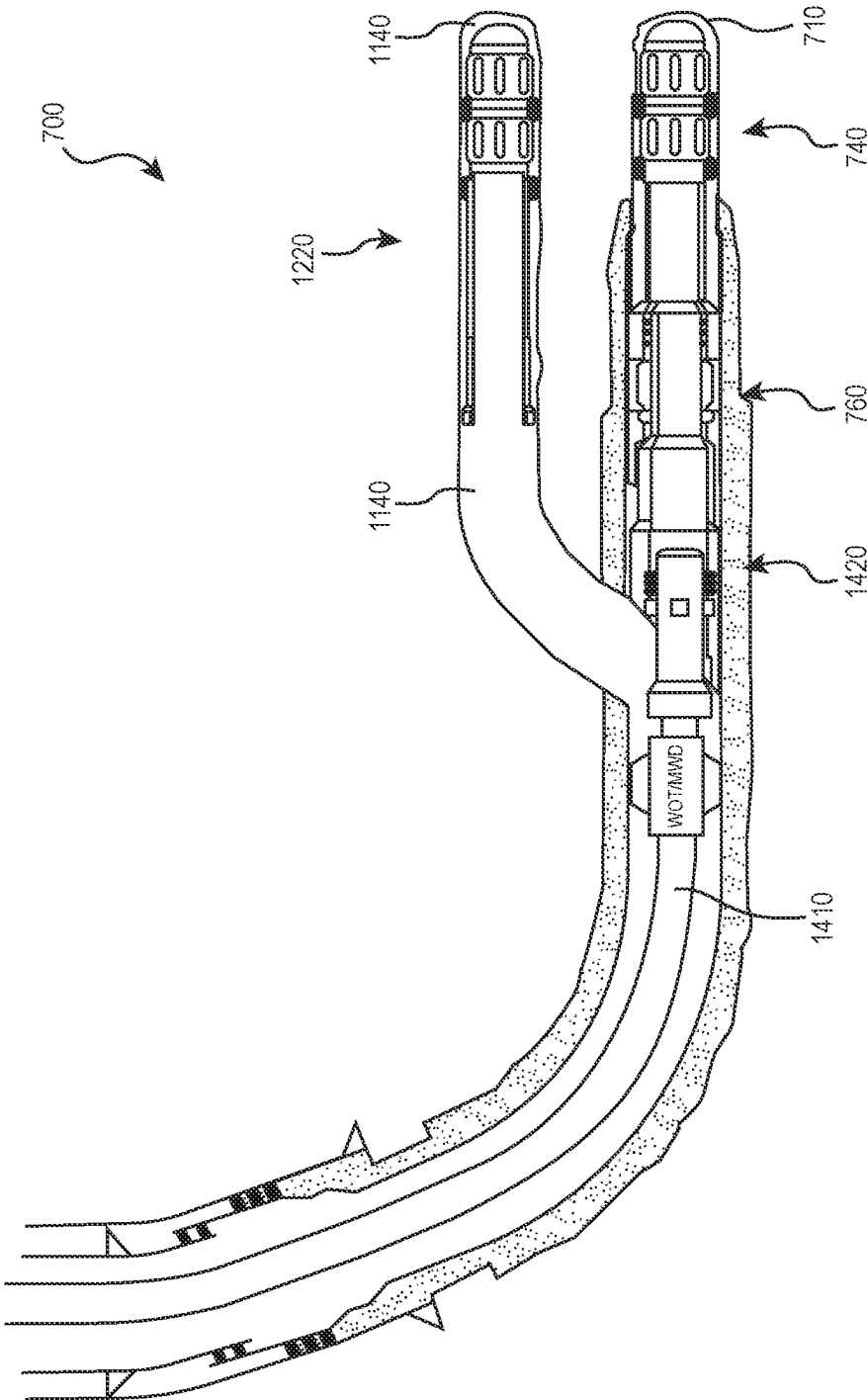
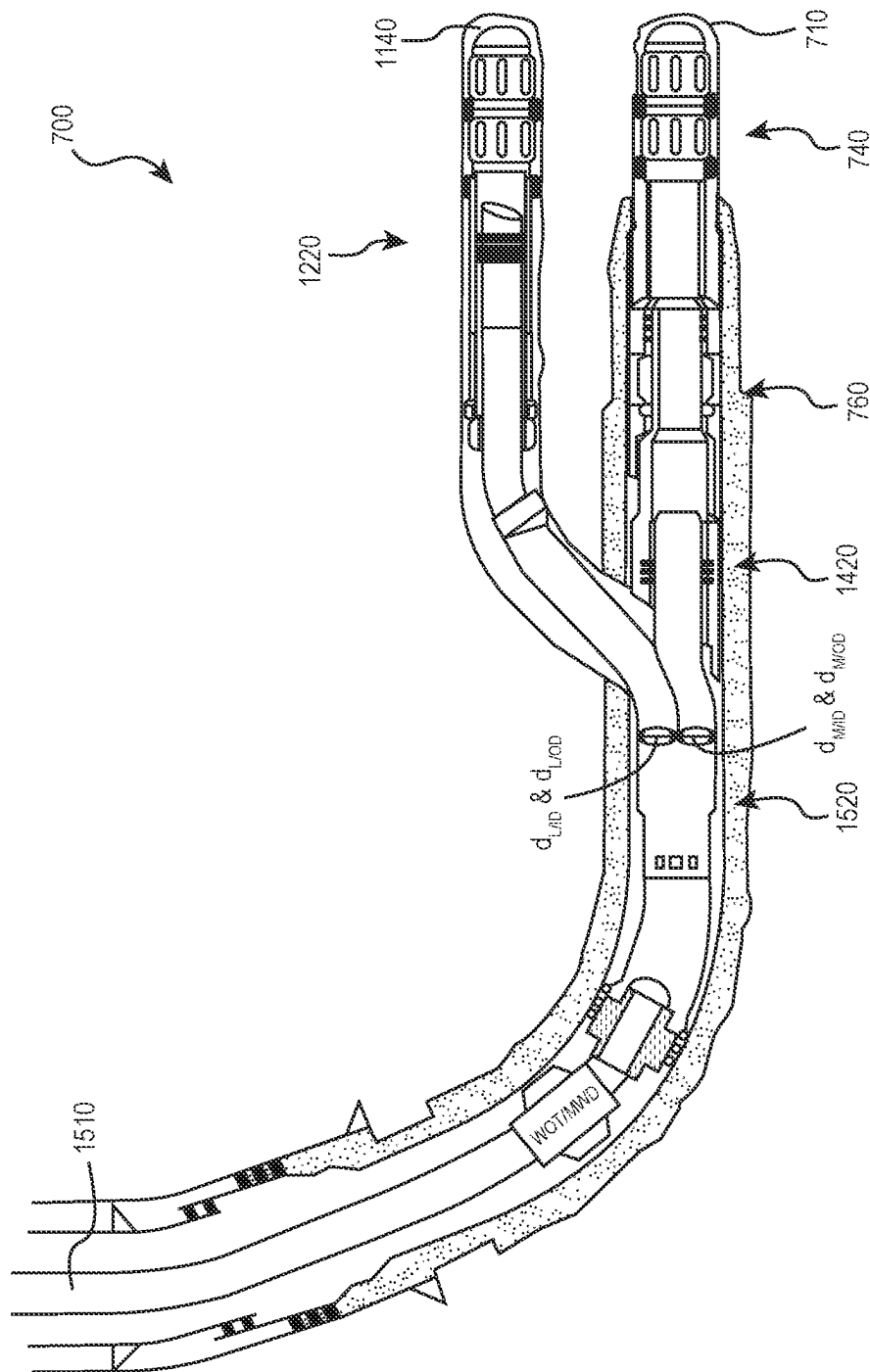


FIG. 14





 DEPARTMENT OF HEALTH AND HUMAN SERVICES

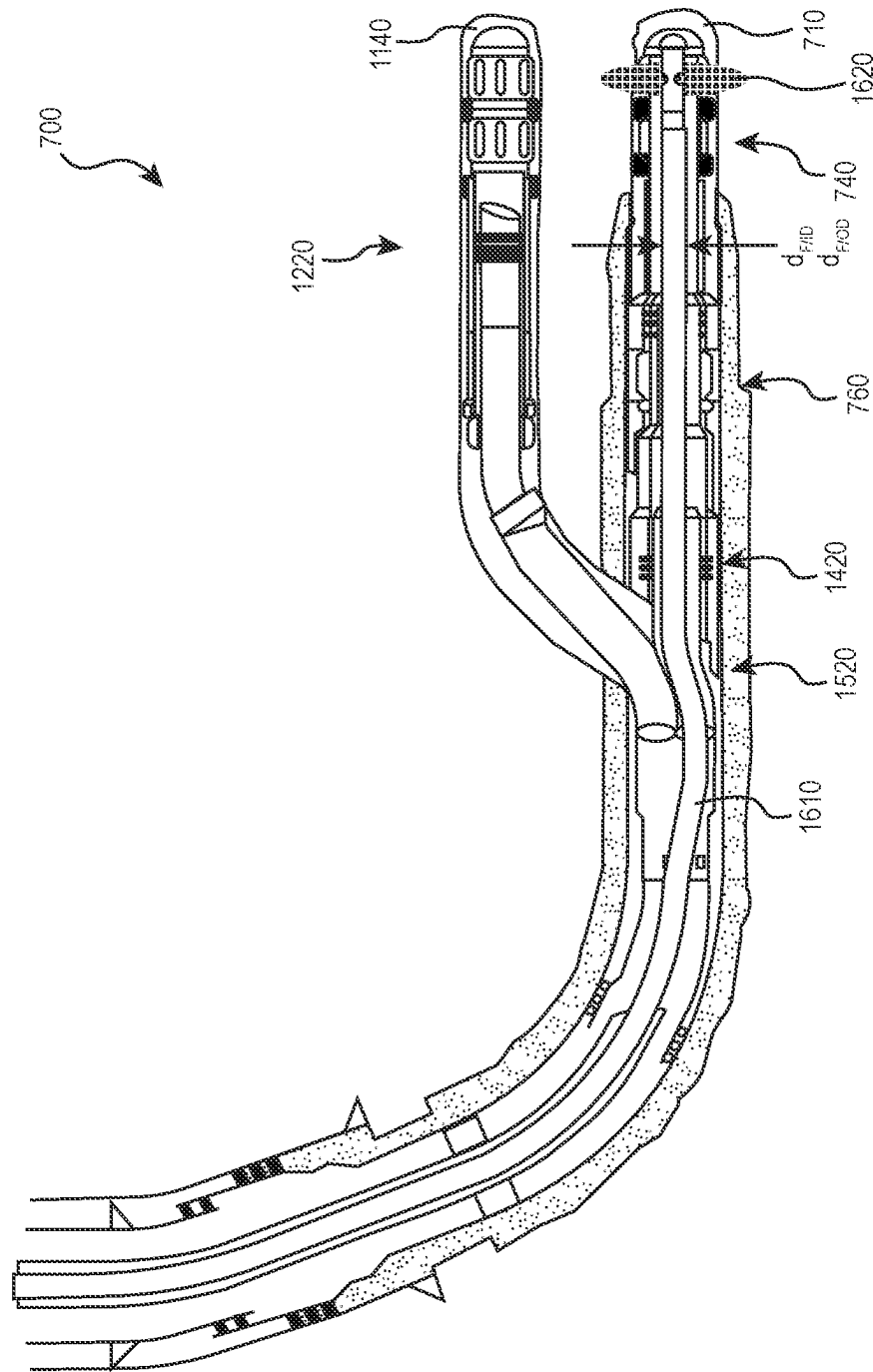


FIG. 16

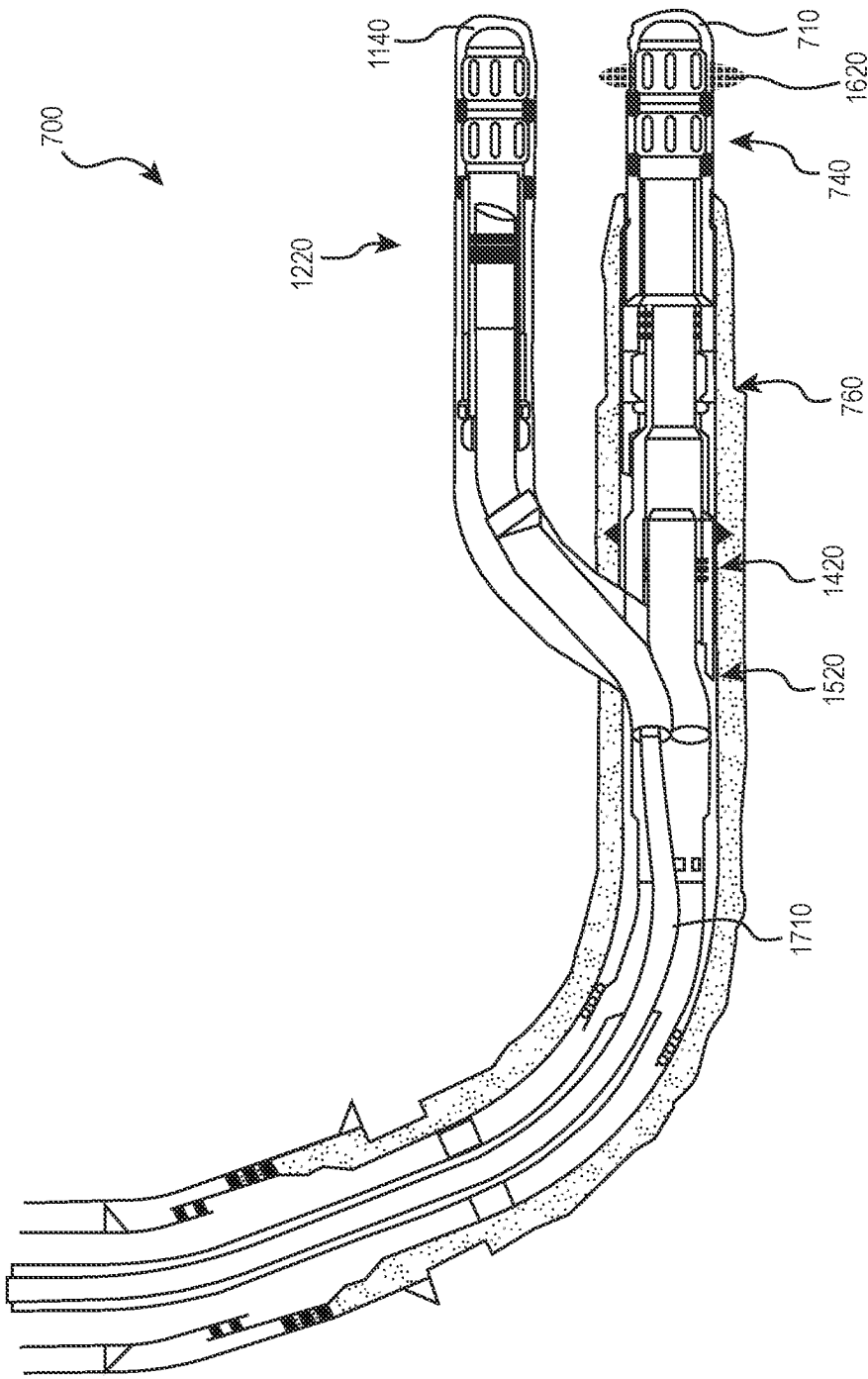


FIG. 17

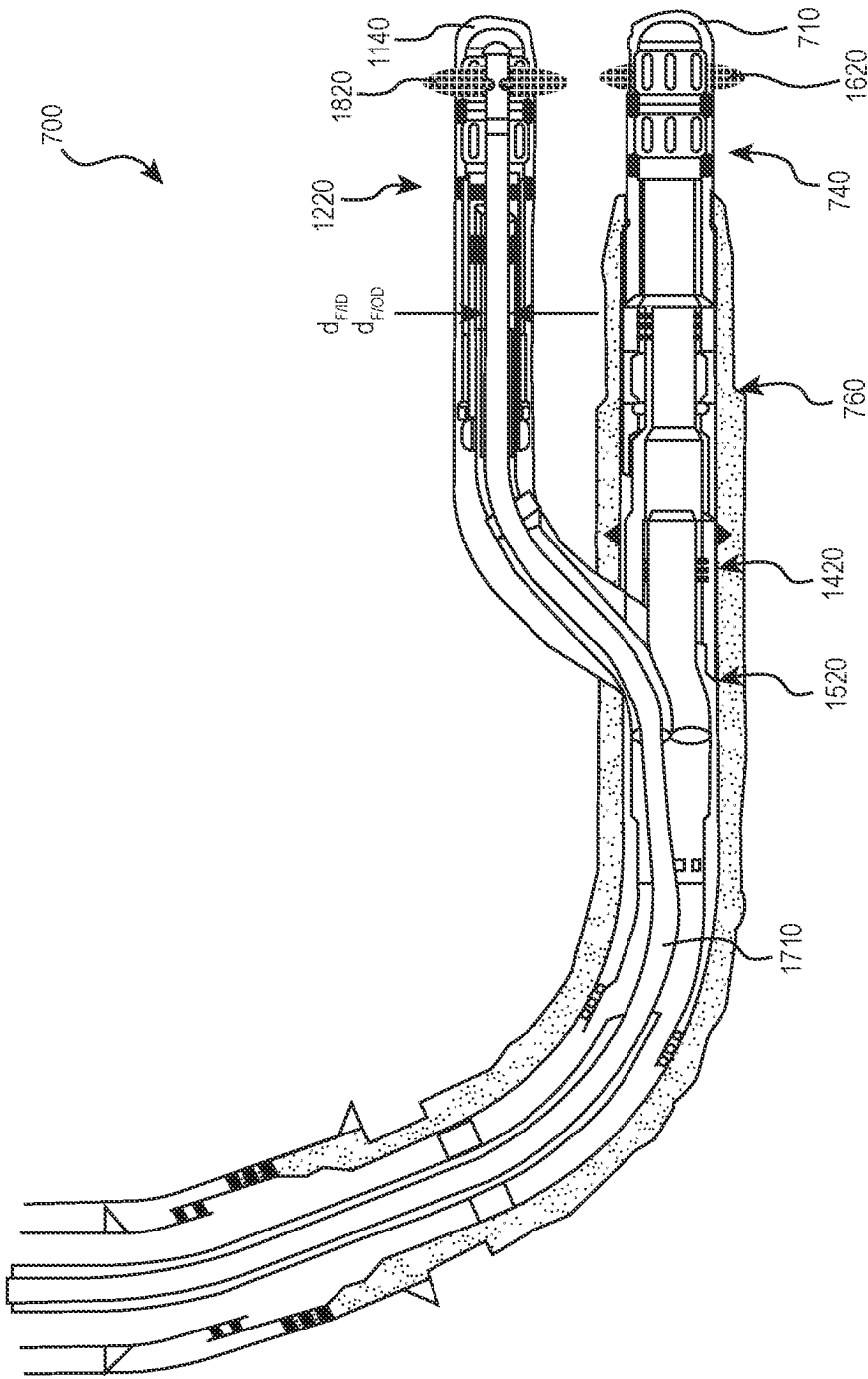


FIG. 18

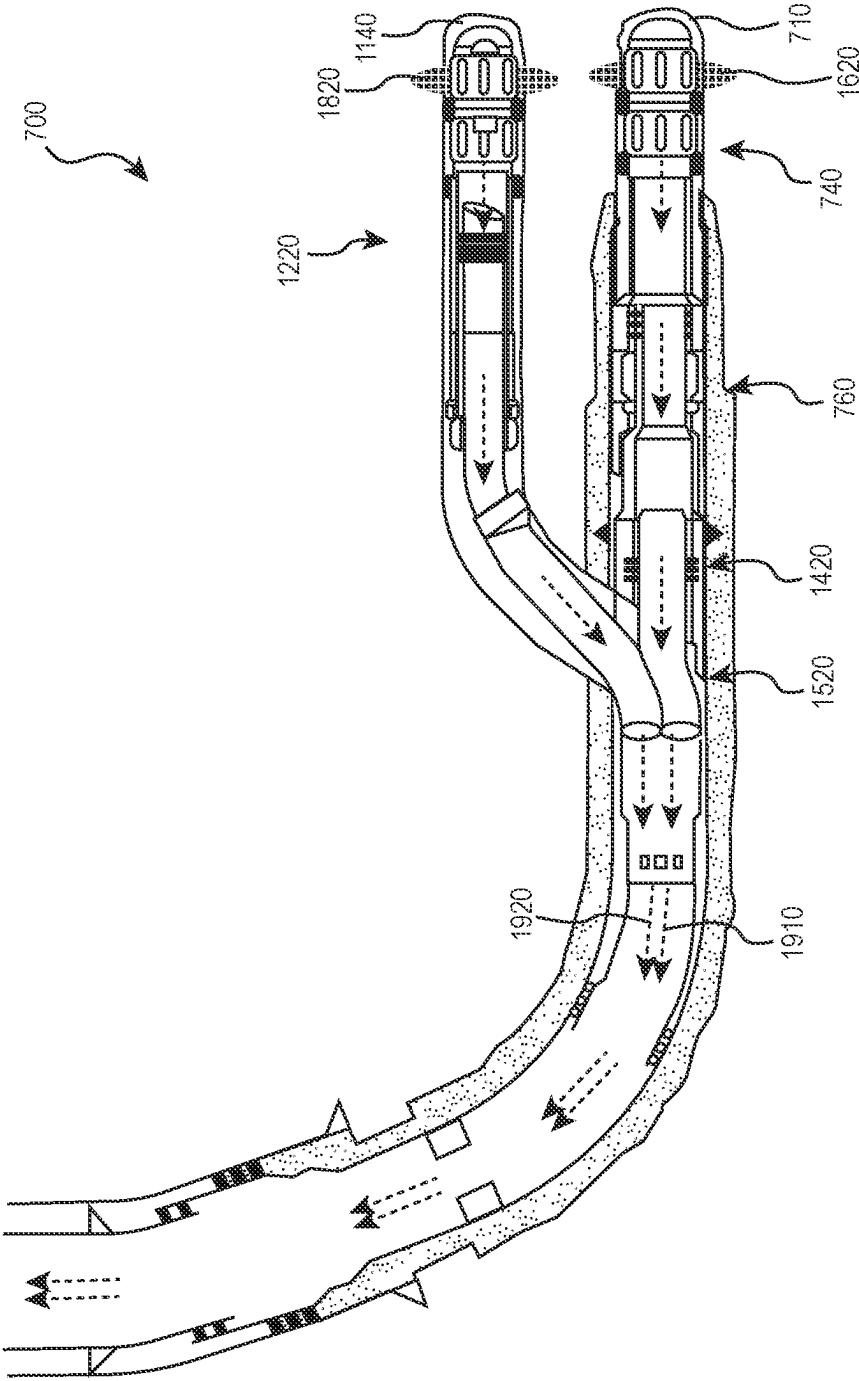


FIG. 19

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HIGH-PRESSURE MULTILATERAL JUNCTION WITH MAINBORE AND LATERAL ACCESS AND CONTROL

CROSS-REFERENCE TO RELATED APPLICATION

This application claims the benefit of U.S. Provisional Application Ser. No. 62/946,219, filed on Dec. 10, 2019, entitled "HIGH PRESSURE MIC WITH MAINBORE AND LATERAL ACCESS AND CONTROL", and incorporated herein by reference in its entirety.

BACKGROUND

A variety of borehole operations require selective access to specific areas of the wellbore. One such selective borehole operation is horizontal multistage hydraulic stimulation, as well as multistage hydraulic fracturing ("frac" or "fracking"). In multilateral wells, the multistage stimulation treatments are performed inside multiple lateral wellbores. Efficient access to all lateral wellbores is critical to complete a successful pressure stimulation treatment, as well as is critical to selectively enter the multiple lateral wellbores with other downhole devices.

BRIEF DESCRIPTION

Reference is now made to the following descriptions taken in conjunction with the accompanying drawings, in which:

FIG. 1 illustrates a well system for hydrocarbon reservoir production, the well system including a y-block designed, manufactured and operated according to one or more embodiments of the disclosure;

FIG. 2 illustrates a perspective view of a y-block designed, manufactured and operated according to one or more embodiments of the disclosure;

FIG. 3 illustrates a cross-section of the perspective view of the y-block illustrated in FIG. 2;

FIG. 4 illustrates a cross-section of a non-perspective view of the y-block illustrated in FIG. 2;

FIGS. 5A and 5B illustrate various different cross-sectional views of an area of the y-block where the second and third bores overlap one another;

FIG. 6 illustrates one embodiment of a multilateral junction designed, manufactured and operated according to the disclosure; and

FIGS. 7 through 19 illustrate a method for forming, fracturing and/or producing from a well system.

DETAILED DESCRIPTION

In the drawings and descriptions that follow, like parts are typically marked throughout the specification and drawings with the same reference numerals, respectively. The drawn figures are not necessarily to scale. Certain features of the disclosure may be shown exaggerated in scale or in somewhat schematic form and some details of certain elements may not be shown in the interest of clarity and conciseness. The present disclosure may be implemented in embodiments of different forms.

Specific embodiments are described in detail and are shown in the drawings, with the understanding that the present disclosure is to be considered an exemplification of the principles of the disclosure, and is not intended to limit the disclosure to that illustrated and described herein. It is to

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be fully recognized that the different teachings of the embodiments discussed herein may be employed separately or in any suitable combination to produce desired results.

Unless otherwise specified, use of the terms "connect," "engage," "couple," "attach," or any other like term describing an interaction between elements is not meant to limit the interaction to a direct interaction between the elements and may also include an indirect interaction between the elements described. Unless otherwise specified, use of the terms "up," "upper," "upward," "uphole," "upstream," or other like terms shall be construed as generally toward the surface of the ground; likewise, use of the terms "down," "lower," "downward," "downhole," or other like terms shall be construed as generally toward the bottom, terminal end of a well, regardless of the wellbore orientation. Use of any one or more of the foregoing terms shall not be construed as denoting positions along a perfectly vertical axis. In some instances, a part near the end of the well can be horizontal or even slightly directed upwards. In such instances, the terms "up," "upper," "upward," "uphole," "upstream," or other like terms shall be used to represent the toward the surface end of a well. Unless otherwise specified, use of the term "subterranean formation" shall be construed as encompassing both areas below exposed earth and areas below earth covered by water such as ocean or fresh water.

A particular challenge for the oil and gas industry is developing a pressure tight TAML (Technology Advancement of Multilaterals) level 5 multilateral junction that can be installed in casing (e.g., 7 $\frac{5}{8}$ " casing) and that also allows for ID access (e.g., ~3 $\frac{1}{2}$ " ID access) to a main wellbore after the junction is installed. This type of multilateral junction could be useful for coiled tubing conveyed stimulation and/or clean-up operations. It is envisioned that future multilateral wells will be drilled from existing slots/wells where additional laterals are added to the existing wellbore. If a side track can be made from the casing (e.g., 9 $\frac{5}{8}$ " casing), there is an option to install a liner (e.g., 7" or 7 $\frac{5}{8}$ " liner) with a new casing exit point positioned at an optimal location to reach undrained reserves.

Referring now to FIG. 1, illustrated is a diagram of a well system 100 for hydrocarbon reservoir production, according to certain example embodiments. The well system 100 in one or more embodiments includes a pumping station 110, a main wellbore 120, tubing 130, 135, which may have differing tubular diameters, and a plurality of multilateral junctions 140, and lateral legs 150 with additional tubing integrated with a main bore of the tubing 130, 135. Each multilateral junction 140 may comprise a junction designed, manufactured or operated according to the disclosure, including a multilateral junction comprising a novel y-block according to the disclosure. The well system 100 may additionally include a control unit 160. The control unit 160, in this embodiment, is operable to control to and/or from the multilateral junctions and/or lateral legs 150, as well as other devices downhole.

Turning to FIG. 2, illustrated is a perspective view of a y-block 200 designed, manufactured and operated according to one or more embodiments of the disclosure. The y-block 200 includes a housing 210. For example, the housing 210 could be a solid piece of metal having been milled to contain various different bores according to the disclosure. In another embodiment, the housing 210 is a cast metal housing formed with the various different bores according to the disclosure. The housing 210, in accordance with one embodiment, may include a first end 220 and a second opposing end 225. The first end 220, in one or more

embodiments, is a first uphole end, and the second end **225**, in one or more embodiments, is a second downhole end.

The housing **210** may have a length (L), which in the disclosed embodiment is defined by the first end **220** and the second opposing end **225**. The length (L) may vary greatly and remain within the scope of the disclosure. In one embodiment, however, the length (L) ranges from about 0.5 meters to about 4 meters. In yet another embodiment, the length (L) ranges from about 1.5 meters to about 2.0 meters, and in yet another embodiment the length (L) is approximately 1.8 meters (e.g., approximately 72 inches).

The y-block **200**, in one or more embodiments, includes a single first bore **230** extending into the housing **210** from the first end **220**. In the disclosed embodiment, the single first bore **230** defines a first centerline **235**. The y-block **200**, in one or more embodiments, further includes a second bore **240** and a third bore **250** extending into the housing **210**. In the illustrated embodiment the second bore **240** and the third bore **250** branch off from the single first bore **230** at a point between the first end **220** and the second opposing end **225**. In accordance with one embodiment of the disclosure, the second bore **240** defines a second centerline **245** and the third bore **250** defines a third centerline **255**. As will be discussed more fully below, the second centerline **245** and the third centerline **255** may be angled relative to one another in one or more embodiments consistent with the disclosure. Moreover, the y-block **200** provides equal and selective access to both legs.

Turning to FIG. 3, illustrated is a cross-section of the perspective view of the y-block **200** illustrated in FIG. 2. FIG. 3 more clearly illustrates the first centerline **235**, the second centerline **245** and the third centerline **255**. FIG. 3 additionally illustrates how the second bore **240** and the third bore **250** branch off from the single first bore **230** at a point between the first end **220** and the second opposing end **225**. Specific to the embodiment of FIG. 3, the second bore **240** and the third bore **250** branch off from the single first bore **230** at a point proximate the first end **220**. In certain embodiments, such as that shown, the second and third bores **240**, **250** overlap one another proximate the single first bore **230**. Accordingly, the overlapped portion of the second and third bores **240**, **250** may provide for commingling of fluids from the second bore **240** and the third bore **250** within the y-block **200**. The second bore **240** and the third bore **250**, in one or more embodiments, are a main leg bore and a lateral leg bore, respectively.

Turning to FIG. 4, illustrated is a cross-section of a non-perspective view of the y-block **200** illustrated in FIG. 2. FIG. 4 further illustrates that the second centerline **245** and the third centerline **255** are angled relative to one another, for example by an angle (β). This angle (β) helps with the frac burst rating of the y-block **200**. In certain embodiments, the angle (β) helps the y-block **200** achieve a 5,000 psi burst rate, and in yet other embodiments the angle (β) helps the y-block **200** achieve a 8,000 psi burst rate, and in even yet other embodiments the angle (β) helps the y-block **200** achieve a 10,000 psi burst rate. The angle (β) may vary greatly based upon the length (L), but tends to be below about 3 degrees. In certain other embodiments, the angle (β) may be less than 2 degrees, and in certain embodiments approximately 1 degree or less.

Further to the embodiment of FIG. 4, the second centerline **245** is angled relative to the first centerline **235**, and in yet another embodiment the third centerline **255** is angled relative to the first centerline **235**. Accordingly, one or both of the second centerline **245** and/or third centerline **255** may be angled relative to the first centerline **235**. For example,

the second centerline **245** might have an angle (θ) between itself and the first centerline **235**, and the third centerline **255** might have an angle (α) between itself and the first centerline **235**. In one or more embodiments of the disclosure, the angle (θ) is greater than the angle (α). The angle (θ) may vary greatly based upon the length (L), but tends to be below about 1 degree. In certain other embodiments, the angle (θ) may be less than 0.75 degrees, and in certain embodiments approximately 0.6 degrees. The angle (α) may also vary greatly based upon the length (L), but tends to be below about 1 degree. In certain other embodiments, the angle (α) may be less than 0.75 degrees, and in certain embodiments approximately 0.4 degrees. Assuming that the first centerline **235** is a horizontal line, and the second centerline **245** angles down from horizontal by the angle (θ) and the third centerline **255** angles up from the horizontal line by the angle (α), the angle (θ) and the angle (α) would add up to the angle (β) discussed above.

The second and third centerlines **245**, **255**, in one or more embodiments, are straight centerlines extending along the length (L) of the y-block **200**. For example, in this embodiment, the second bore **240** and/or third bore **250** would each include only a single straight centerline, as opposed to each including two or more angularly offset centerlines. Moreover, even if the second bore **240** and/or third bore **250** have different portions with different diameters, such as is the case with the second bore **240** illustrated in FIG. 4, in accordance with this embodiment the second centerline **245** and third centerline **255** would still each include only a single straight centerline. Moreover, it should be noted that centerline **235** does not need to be concentric with the OD defined by **244**.

The single first bore **230**, the second bore **240** and the third bore **250** may have different diameters and remain within the scope of the disclosure. In one embodiment, the single first bore **230** has a diameter (d_1). The diameter (d_1) may range greatly, but in one or more embodiments the diameter (d_1) ranges from about 2.5 cm to about 60.1 cm (e.g., from about 1 inches to about 24 inches). The diameter (d_1), in one or more embodiments, ranges from about 7.6 cm to about 40.6 cm (e.g., from about 3 inches to about 16 inches). In yet another embodiment, the diameter (d_1) may range from about 15.2 cm to about 30.5 cm (e.g., from about 6 inches to about 12 inches). In yet another embodiment, the diameter (d_1) may range from about 17.8 cm to about 25.4 cm (e.g., from about 7 inches to about 10 inches), and more specifically in one embodiment a value of about 21.6 cm (e.g., about 8.5 inches).

In one embodiment, the third bore **250** has a diameter (d_3). The diameter (d_3) may range greatly, but in one or more embodiments the diameter (d_3) ranges from about 0.64 cm to about 50.8 cm (e.g., from about 1/4 inches to about 20 inches). The diameter (d_3), in one or more other embodiments, ranges from about 2.5 cm to about 17.8 cm (e.g., from about 1 inches to about 7 inches). In yet another embodiment, the diameter (d_3) may range from about 6.4 cm to about 12.7 cm (e.g., from about 2.5 inches to about 5 inches). In yet another embodiment, the diameter (d_3) may range from about 7.6 cm to about 10.2 cm (e.g., from about 3 inches to about 4 inches), and more specifically in one embodiment a value of about 8.9 cm (e.g., about 3.5 inches).

In one embodiment, the second bore **240** has a diameter (d_2). The diameter (d_2) may range greatly, but in one or more embodiments the diameter (d_2) ranges from about 0.64 cm to about 50.8 cm (e.g., from about 1/4 inches to about 20 inches). The diameter (d_2), in one or more embodiments, ranges from about 2.5 cm to about 17.8 cm (e.g., from about

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1 inches to about 7 inches). In yet another embodiment, the diameter (d_2) may range from about 6.4 cm to about 12.7 cm (e.g., from about 2.5 inches to about 5 inches). In yet another embodiment, the diameter (d_2) may range from about 7.6 cm to about 10.2 cm (e.g., from about 3 inches to about 4 inches), and more specifically in one embodiment a value of about 8.9 cm (e.g., about 3.5 inches). In certain other embodiments, the second bore **240** has a first portion **242** and a second portion **244**. In the illustrated embodiment, the first portion **242** has the diameter (d_2) and the second portion has a greater diameter (d_2). The greater diameter (d_2) provides the ability to land all the necessary tools within the y-block **200**. The greater diameter (d_2) may range greatly, but in one or more embodiments the greater diameter (d_2) ranges from about 0.95 cm to about 53.3 cm (e.g., from about $\frac{3}{8}$ inches to about 21 inches). The greater diameter (d_2), in one or more embodiments, ranges from about 3.18 cm to about 18.4 cm (e.g., from about 1.25 inches to about 7.25 inches). In yet another embodiment, the greater diameter (d_2) may range from about 7 cm to about 13.34 cm (e.g., from about 2.75 inches to about 5.25 inches). In yet another embodiment, the diameter (d_2) may range from about 8.26 cm to about 10.8 cm (e.g., from about 3.25 inches to about 4.25 inches), and more specifically in one embodiment a value of about 9.53 cm (e.g., about 3.75 inches). In certain other embodiments, the second diameter (d_2) is equal to the third diameter (d_3), and the greater diameter (d_2) is larger than both the second diameter (d_2) and the third diameter (d_3).

In certain embodiments, the second portion **244** is located between the first portion **242** and the single first bore **230**. Furthermore, in certain other embodiments, the first portion **242** has a length (L_1) and the second portion **244** has a length (L_2). In accordance with one or more embodiments, the length (L_2) of the second portion **244** is at least two times a length (L_1) of the first portion **242**. In accordance with one or more other embodiments, the length (L_2) of the second portion **244** is at least three times a length (L_1) of the first portion **242**.

The single first bore **230**, second bore **240** and third bore **250**, in one or more embodiments, are configured to connect with various different features. For example, in one or more embodiments, the single first bore **230** may include a box joint or a pin joint for engaging with the other uphole features. Similarly, the second bore **240** could include a box joint or a pin joint for engaging with the other downhole features, such as main wellbore leg. In one or more other embodiments, the third bore **250** might be relegated to a box joint for engaging with other downhole features, such as the lateral wellbore leg. Nevertheless, the present disclosure should not limit the type of joint any of the single first bore **230**, second bore **240** or third bore **250** could employ.

Turning to FIGS. **5A** and **5B**, illustrated are various different cross-sectional views of an area of the y-block **200** where the second and third bores **240**, **250** overlap one another. As shown in FIG. **5A**, which is similar to the y-block of FIG. **4**, a shared interior wall **510** of the second and third bores **240**, **250** comes to a blunt stress relief point **520** at a location wherein the second and third bores **240**, **250** come together. Essentially, the blunt stress relief point **520** removes the extremely thin sidewall areas of the second and third bores **240**, **250** as they approach one another, for example to prevent them from physically collapsing under pressure and potentially damaging the y-block **200**. This blunt stress relief point **520**, may be formed by one or more different machining processes. In certain embodiments, a milling process is used to form the blunt stress relief point

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520. In other embodiments, an electric discharge machining (EDM) process is used to form the blunt stress relief point **520**. In contrast, FIG. **5B** illustrates the shared interior wall of the second and third bores **240**, **250** coming to a sharp point **530** at a location wherein the second and third bores **240**, **250** come together. The sharp point **530** remains susceptible to collapsing, but in certain embodiments the collapsed area adds certain benefits.

Turning to FIG. **6**, illustrated is one embodiment of a multilateral junction **600** designed, manufactured and operated according to the disclosure. The multilateral junction **600**, in the illustrated embodiment, includes a y-block **610**, a main bore leg **620**, and a lateral bore leg **630**. The y-block **610** may comprise any y-block consistent with the disclosure, including the y-block **200** discussed above with regard to FIGS. **2** through **5B**.

The main bore leg **620** and the lateral bore leg **630**, in the illustrated embodiment, are threadingly engaged with the y-block **610**. In at least one or more embodiments, the main bore leg **620** includes an outer diameter ($d_{M(OD)}$) and an inner diameter ($d_{M(ID)}$). Similarly, the lateral bore leg **630** includes an outer diameter ($d_{L(OD)}$) and an inner diameter ($d_{L(ID)}$).

The multilateral junction **600** may additionally include a tubular **640**. The tubular **640**, in at least one embodiment, is configured to provide a consistent (e.g., laminar) flow through the multilateral junction **600** to reduce turbulence, and include a spacer insert. The tubular **640** may also, in certain embodiments, guide an intervention tool (e.g., frac tool) into the second bore (e.g., main bore) side of the y-block **610**.

In accordance with one embodiment, a deflector (not shown) may be installed in or above the y-block **610**. The deflector, in this example, may be permanently installed, run-in-hole on a separate trip as the multilateral junction **600**, or run-in-hole on the same trip as the multilateral junction **600**. In other embodiments, the y-block **610** is a single solid housing having the single first bore, the second bore and the third bore formed therein, and to the extent a deflector is necessary, it is formed as an integral portion of the housing. This is opposed to a situation where a separate deflector assembly is positioned within the housing of the y-block **610**.

The multilateral junction **600** may additionally include a TEW ("tubing exit whipstock) sleeve. The TEW sleeve, not shown, is located within the tubular **640** and in certain embodiments proximate to or within the y-block **610**. The TEW sleeve, when used, is operable to deflect intervention tools (e.g., such as a fracturing string), into the third bore (e.g., lateral bore) of the multilateral junction **600**. The TEW sleeve may be installed and retrieved using a hydraulic running/retrieving tool, among other tools. In certain embodiments, the TEW sleeve is held in the multilateral junction **600** using a collet.

The multilateral junction **600**, in one or more embodiments, is a high pressure multilateral junction. For example, in at least one embodiment, the multilateral junction **600** is capable of withstanding at least 8,000 psi burst rate. In yet another example, the multilateral junction **600** is capable of withstanding at least 10,000 psi burst rate. In at least one embodiment, the multilateral junction **600** is capable of withstanding at least 5000 psi collapse rate. In yet another example, the multilateral junction **600** is capable of withstanding at least 7000 psi collapse rate. Accordingly, the multilateral junction **600** may be employed to access and fracture one or both of the main wellbore and/or lateral wellbore. For example, the multilateral junction **600** could have the necessary pressure ratings, outside diameters, and

inside diameters necessary to run a fracturing string there through, and thereafter appropriately and safely fracture one or both of the main wellbore and/or lateral wellbore.

Thus, in accordance with one or more embodiments of the disclosure, the multilateral junction **600** is capable of with-
standing at least 10,000 psi burst rate, with the main bore leg **620** and lateral bore leg **630** having an inner diameter ($d_{M/ID}$) and diameter ($d_{L/ID}$), respectively, of at least about 80 mm (e.g., about 3.15 inches). In certain other embodiments, the main bore leg **620** and lateral bore leg **630** have an inner diameter ($d_{M/ID}$) and diameter ($d_{L/ID}$), respectively, of at least about 87 mm (e.g., about 3.423 inches). In certain other embodiments, the main bore leg **620** and lateral bore leg **630** have an inner diameter ($d_{M/ID}$) and diameter ($d_{L/ID}$), respectively, of at least about 90 mm (e.g., about 3.548 inches). In certain embodiments, the main bore leg **620** and lateral bore leg **630** have an outer diameter ($d_{M/OD}$) and diameter ($d_{L/OD}$), respectively, of at least about 101.6 mm (e.g., about 4.0 inches). Accordingly, a fracturing string having an outside diameter ($d_{F/OD}$) of at least about 78 mm (e.g., about 3.07 inches) could travel through the multilateral junction **600** and engage with a main wellbore completion or lateral wellbore completion, for fracturing the main wellbore or lateral wellbore, respectively. In yet another embodiment, a fracturing string having an outside diameter ($d_{F/OD}$) of at least about 85.7 mm (e.g., about 3.375 inches) could travel through the multilateral junction **600** and engage with a main wellbore completion or lateral wellbore completion, for fracturing the main wellbore or lateral wellbore, respectively. Such fracturing strings may additionally have an inside diameter ($d_{J/ID}$) of at least about 50.8 mm (e.g., about 2 inches). In accordance with this embodiment, the diameter (d_2) of the second bore of the y-block **610** and the diameter (d_3) of the third bore of the y-block **610**, might have an inside diameter, of about 87 mm (e.g., about 3.423 inches). Heretofore, nobody was capable of fracturing through the main bore leg **620** and/or lateral bore leg **630** at the aforementioned high pressures.

Turning now to FIGS. **7** through **18**, illustrated is a method for forming, accessing, potentially fracturing, and producing from a well system **700**. FIG. **7** is a schematic of the well system **700** at the initial stages of formation. A main wellbore **710** may be drilled, for example by a rotary steerable system at the end of a drill string and may extend from a well origin (not shown), such as the earth's surface or a sea bottom. The main wellbore **710** may be lined by one or more casings **715**, **720**, each of which may be terminated by a shoe **725**, **730**.

The well system **700** of FIG. **7** additionally includes a main wellbore completion **740** positioned in the main wellbore **710**. The main wellbore completion **740** may, in certain embodiments, include a main wellbore liner **745** (e.g., with frac sleeves in one embodiment), as well as one or more packers **750** (e.g., swell packers in one embodiment). The main wellbore liner **745** and the one or more packer **750** may, in certain embodiments, be run on an anchor system **760**. The anchor system **760**, in one embodiment, includes a collet profile **765** for engaging with the running tool **790**, as well as a muleshoe **770** (e.g., slotted alignment muleshoe). A standard workstring orientation tool (WOT) and measurement while drilling (MWD) tool may be coupled to the running tool **790**, and thus be used to orient the anchor system **760**.

Turning to FIG. **8**, illustrated is the well system **700** of FIG. **7** after positioning a whipstock assembly **810** downhole at a location where a lateral wellbore is to be formed. The whipstock assembly **810** includes a collet **820** for

engaging the collet profile **765** in the anchor system **760**. The whipstock assembly **810** additionally includes one or more seals **830** (e.g., a wiper set in one embodiment) to seal the whipstock assembly **810** with the main wellbore completion **740**. In certain embodiments, such as that shown in FIG. **8**, the whipstock assembly **810** is made up with a lead mill **840**, for example using a shear bolt, and then run in hole on a drill string **850**. The WOT/MWD tool may be employed to orient the whipstock assembly **810**.

Turning to FIG. **9**, illustrated is the well system **700** of FIG. **8** after setting down weight to shear the shear bolt between the lead mill **840** and the whipstock assembly **810**, and then milling an initial window pocket **910**. In certain embodiments, the initial window pocket **910** is between 1.5 m and 7.0 m long, and in certain other embodiments about 2.5 m long, and extends through the casing **720**. Thereafter, a circulate and clean process could occur, and then the drill string **850** and lead mill **840** may be pulled out of hole.

Turning to FIG. **10**, illustrated is the well system **700** of FIG. **9** after running a lead mill **1020** and watermelon mill **1030** downhole on a drill string **1010**. In the embodiments shown in FIG. **10**, the drill string **1010**, lead mill **1020** and watermelon mill **1030** drill a full window pocket **1040** in the formation. In certain embodiments, the full window pocket **1040** is between 5 m and 10 m long, and in certain other embodiments about 8.5 m long. Thereafter, a circulate and clean process could occur, and then the drill string **1010**, lead mill **1020** and watermelon mill **1030** may be pulled out of hole.

Turning to FIG. **11**, illustrated is the well system **700** of FIG. **10** after running in hole a drill string **1110** with a rotary steerable assembly **1120**, drilling a tangent **1130** following an inclination of the whipstock assembly **810**, and then continuing to drill the lateral wellbore **1140** to depth. Thereafter, the drill string **1110** and rotary steerable assembly **1120** may be pulled out of hole.

Turning to FIG. **12**, illustrated is the well system **700** of FIG. **11** after employing an inner string **1210** to position a lateral wellbore completion **1220** in the lateral wellbore **1140**. The lateral wellbore completion **1220** may, in certain embodiments, include a lateral wellbore liner **1230** (e.g., with frac sleeves in one embodiment), as well as one or more packers **1240** (e.g., swell packers in one embodiment). Thereafter, the inner string **1210** may be pulled into the main wellbore **710** for retrieval of the whipstock assembly **810**.

Turning to FIG. **13**, illustrated is the well system **700** of FIG. **12** after latching a whipstock retrieval tool **1310** of the inner string **1210** with a profile in the whipstock assembly **810**. The whipstock assembly **810** may then be pulled free from the anchor system **760**, and then pulled out of hole. What results are the main wellbore completion **740** in the main wellbore **710**, and the lateral wellbore completion **1220** in the lateral wellbore **1140**.

Turning to FIG. **14**, illustrated is the well system **700** of FIG. **13** after employing a running tool **1410** to install a deflector assembly **1420** proximate a junction between the main wellbore **710** and the lateral wellbore **1140**. The deflector assembly **1420** may be appropriately oriented using the WOT/MWD tool. The running tool **1410** may then be pulled out of hole.

Turning to FIG. **15**, illustrated is the well system **700** of FIG. **14** after employing a running tool **1510** to place a multilateral junction **1520** proximate an intersection between the main wellbore **710** and the lateral wellbore **1140**. In accordance with one embodiment, the multilateral junction **1520** may include similar features as the multilateral junction **600** discussed above. Accordingly, the multi-

lateral junction **1520** may be installed as a unitary junction, wherein the y-block, mainbore leg and lateral bore leg are all run at the same time. In another embodiments, other types of multilateral junctions **1520** maybe utilized, such as a two-piece junction where a portion of the multilateral junction (e.g., the mainbore leg) is run separately prior to running of the other portion of the junction (e.g., lateral bore leg). In other embodiments, where large-access to the mainbore and/or lateral leg is not required, a multilateral junction **1520** with smaller legs may be used. Accordingly, the multilateral junction **1520** would include a y-block designed, manufactured, and operated according to one or more embodiments of the disclosure, and could be operable to handle at least 8,000 psi burst rate, or in yet another embodiment at least about 10,000 psi burst rate.

In the illustrated embodiment, the multilateral junction **1520** includes a y-block similar to the y-block **200** illustrated with respect to FIGS. **2** through **5B**. For example, while not easily illustrated given the scale of FIG. **15**, the multilateral junction **1520** could have a y-block with the aforementioned second and third centerlines that are angled relative to one another. Moreover, the main bore leg and lateral bore leg could have the inner diameters ($d_{M/ID}$) and diameters ($d_{L/ID}$), as well as the outer diameters ($d_{M/OD}$) and diameters ($d_{L/OD}$) discussed above. For example, the main bore leg and the lateral bore leg might have an inner diameter ($d_{M/ID}$) and diameter ($d_{L/ID}$) of at least about 80 mm, or in another embodiment of at least about 87 mm, or in yet another embodiment of at least about 90 mm.

Turning to FIG. **16**, illustrated is the well system **700** of FIG. **15** after selectively accessing the main wellbore **710** with a first intervention tool **1610** through the y-block of the multilateral junction **1520**. In the illustrated embodiment, the first intervention tool **1610** is a first fracturing string, and more particularly a coiled tubing conveyed fracturing string. The first fracturing string may have any of the outside diameters ($d_{F/OD}$) and inside diameters ($d_{F/ID}$) discussed above and remain within the scope of the disclosure. For example, the first fracturing string could have an outside diameter ($d_{F/OD}$) of at least about 78 mm, or in yet another embodiment of at least about 85.7. Similarly, the first fracturing string could have an inside diameter ($d_{F/ID}$) of at least about 50.8 mm. With the first intervention tool **1610** in place, fractures **1620** in the subterranean formation surrounding the main wellbore completion **740** may be formed. Thereafter, the first intervention tool **1610** may be pulled from the main wellbore completion **740**.

Turning to FIG. **17**, illustrated is the well system **700** of FIG. **16** after positioning a second intervention tool **1710** within the multilateral junction **1520** including the y-block. In the illustrated embodiment, the second intervention tool **1710** is a second fracturing string, and more particularly a coiled tubing conveyed fracturing string. The second fracturing string may have any of the outside diameters ($d_{F/OD}$) and inside diameters ($d_{F/ID}$) discussed above and remain within the scope of the disclosure. For example, the second fracturing string could have an outside diameter ($d_{F/OD}$) of at least about 78 mm, or in yet another embodiment of at least about 85.7. Similarly, the second fracturing string could have an inside diameter ($d_{F/ID}$) of at least about 50.8 mm.

Turning to FIG. **18**, illustrated is the well system **700** of FIG. **17** after putting additional weight down on the second intervention tool **1710** and causing the second intervention tool **1710** to enter the lateral wellbore **1140**. With the downhole tool **1710** in place, fractures **1820** in the subterranean formation surrounding the lateral wellbore completion **1220** may be formed. In certain embodiments, the first

intervention tool **1610** and the second intervention tool **1710** are the same intervention tool, and thus the same fracturing tool in one or more embodiments. Thereafter, the second intervention tool **1710** may be pulled from the lateral wellbore completion **1220** and out of the hole.

The embodiments discussed above reference that the main wellbore **710** is selectively accessed and fractured prior to the lateral wellbore **1140**. Nevertheless, other embodiments may exist wherein the lateral wellbore **1140** is selectively accessed and fractured prior to the main wellbore **710**. The embodiments discussed above additionally reference that both the main wellbore **710** and the lateral wellbore **1140** are selectively accessed and fractured through the y-block. Other embodiments may exist wherein only one of the main wellbore **710** or the lateral wellbore **1140** is selectively accessed and fractured through the y-block.

Turning to FIG. **19**, illustrated is the well system **700** of FIG. **18** after producing fluids **1910** from the fractures **1620** in the main wellbore **710**, and producing fluids **1920** from the fractures **1820** in the lateral wellbore **1140**. The producing of the fluids **1910**, **1920** occur through the multilateral junction **1520**, and more specifically through the y-block design, manufactured and operated according to one or more embodiments of the disclosure.

Aspects disclosed herein include:

A. A y-block, the y-block including: 1) a housing having a first end and a second opposing end; 2) a single first bore extending into the housing from the first end, the single first bore defining a first centerline; and 3) second and third separate bores extending into the housing and branching off from the single first bore, the second bore defining a second centerline and the third bore defining a third centerline, wherein the second and third centerlines are angled relative to one another.

B. A multilateral junction, the multilateral junction including: 1) a y-block, the y-block including; a) a housing having a first end and a second opposing end; b) a single first bore extending into the housing from the first end, the single first bore defining a first centerline; and c) second and third separate bores extending into the housing and branching off from the single first bore, the second bore defining a second centerline and the third bore defining a third centerline, wherein the second and third centerlines are angled relative to one another; 2) a mainbore leg coupled to the second bore for extending into the main wellbore; and 3) a lateral bore leg coupled to the third bore for extending into the lateral wellbore.

C. A well system, the well system including: 1) a main wellbore; 2) a lateral wellbore extending from the main wellbore; and 3) a multilateral junction positioned at an intersection of the main wellbore and the lateral wellbore, the multilateral junction including; a) a y-block, the y-block including; i) a housing having a first end and a second opposing end; ii) a single first bore extending into the housing from the first end, the single first bore defining a first centerline; and iii) second and third separate bores extending into the housing and branching off from the single first bore, the second bore defining a second centerline and the third bore defining a third centerline, wherein the second and third centerlines are angled relative to one another; b) a mainbore leg coupled to the second bore and extending into the main wellbore; and c) a lateral bore leg coupled to the third bore and extending into the lateral wellbore.

D. A method for accessing a well system, the method including: 1) placing a multilateral junction proximate an intersection between a main wellbore and a lateral wellbore, the multilateral junction including; a) a y-block, the y-block

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including: i) a housing having a first end and a second opposing end; ii) a single first bore extending into the housing from the first end, the single first bore defining a first centerline; and iii) second and third separate bores extending into the housing and branching off from the single first bore, the second bore defining a second centerline and the third bore defining a third centerline; b) a mainbore leg coupled to the second bore and extending into the main wellbore; and c) a lateral bore leg coupled to the third bore and extending into the lateral wellbore; and 2) selectively accessing at least one of the main wellbore or the lateral wellbore with a fracturing string through the y-block.

Aspects A, B, C and D may have one or more of the following additional elements in combination: Element 1: wherein the second centerline is angled relative to the first centerline. Element 2: wherein the third centerline is angled relative to the first centerline. Element 3: wherein the second centerline has a greater angle (θ) between itself and the first centerline than an angle (α) between the third centerline and the first centerline. Element 4: wherein the second bore is a main leg bore and the third bore is a lateral leg bore. Element 5: wherein the second bore has a diameter (d_2) and the third bore has a diameter (d_3), and further wherein the diameter (d_2) is the same as the diameter (d_3). Element 6: wherein the second bore has a first portion having the diameter (d_2) and a second portion having a greater diameter ($d_{2'}$). Element 7: wherein the second portion is located between the first portion and the single first bore. Element 8: wherein a length (L_2) of the second portion is at least two times a length (L_1) of the first portion. Element 9: wherein the second and third bores overlap one another proximate the single first bore. Element 10: wherein a shared interior wall of the second and third bores comes to a sharp point at a location wherein the second and third bores overlap one another. Element 11: wherein a shared interior wall of the second and third bores comes to a blunt stress relief point at a location wherein the second and third bores overlap one another. Element 12: wherein the third bore includes a box joint at the second opposing end. Element 13: wherein the second bore includes a pin joint at the second opposing end. Element 14: wherein the mainbore leg and the lateral bore leg are threadingly engaged with the y-block. Element 15: wherein the second bore and the third bore each include only a single straight centerline. Element 16: wherein the second centerline is angled relative to the first centerline. Element 17: wherein the third centerline is angled relative to the first centerline. Element 18: wherein the second centerline has a greater angle (θ) between itself and the first centerline than an angle (α) between the third centerline and the first centerline. Element 19: wherein the second bore is a main leg bore and the third bore is a lateral leg bore. Element 20: wherein the second bore has a diameter (d_2) and the third bore has a diameter (d_3), and further wherein the diameter (d_2) is the same as the diameter (d_3). Element 22: wherein the second bore has a first portion having the diameter (d_2) and a second portion having a greater diameter ($d_{2'}$). Element 23: wherein the second portion is located between the first portion and the single first bore. Element 24: wherein a length (L_2) of the second portion is at least two times a length (L_1) of the first portion. Element 25: wherein the second and third bores overlap one another proximate the single first bore. Element 26: wherein a shared interior wall of the second and third bores comes to a sharp point at a location wherein the second and third bores overlap one another. Element 27: wherein a shared interior wall of the second and third bores comes to a blunt stress relief point at a location wherein the second and third bores overlap one another. Element 28: wherein the

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third bore includes a box joint at the second opposing end. Element 29: wherein the second bore includes a pin joint at the second opposing end. Element 30: wherein the mainbore leg and the lateral bore leg are threadingly engaged with the y-block. Element 31: wherein the second bore and the third bore each include only a single straight centerline. Element 32: wherein the second centerline is angled relative to the first centerline and the third centerline is angled relative to the first centerline. Element 33: wherein the second centerline has a greater angle (θ) between itself and the first centerline than an angle (α) between the third centerline and the first centerline. Element 34: wherein the second bore has a first portion having the diameter (d_2) and a second portion having a greater diameter ($d_{2'}$). Element 35: wherein the second portion is located between the first portion and the single first bore. Element 36: wherein the second and third bores overlap one another proximate the single first bore. Element 37: wherein a shared interior wall of the second and third bores comes to a sharp point at a location wherein the second and third bores overlap one another. Element 38: wherein a shared interior wall of the second and third bores comes to a blunt stress relief point at a location wherein the second and third bores overlap one another. Element 39: wherein further including fracturing the at least one of the main wellbore or the lateral wellbore with the fracturing string extending through the y-block. Element 40: wherein selectively accessing at least one of the main wellbore or the lateral wellbore includes selectively accessing the main wellbore with a first fracturing string through the y-block. Element 41: further including fracturing the main wellbore with the first fracturing string extending through the y-block. Element 42: further including selectively accessing the lateral wellbore with a second fracturing string through the y-block. Element 43: further including fracturing the lateral wellbore with the second fracturing string extending through the y-block. Element 44: wherein selectively accessing the main wellbore and fracturing the main wellbore occurs prior to selectively accessing the lateral wellbore and fracturing the lateral wellbore. Element 45: wherein selectively accessing the main wellbore and fracturing the main wellbore occurs after selectively accessing the lateral wellbore and fracturing the lateral wellbore. Element 46: wherein further including producing fluids from fractures in the main wellbore and fractures in the lateral wellbore through the y-block. Element 47: wherein the main bore leg and lateral bore leg have an inner diameter (d_{MID}) and diameter (d_{LID}), respectively, of at least about 80 mm. Element 48: wherein the main bore leg and lateral bore leg have an inner diameter (d_{MID}) and diameter (d_{LID}), respectively, of at least about 87 mm. Element 49: wherein the main bore leg and lateral bore leg have an inner diameter (d_{MID}) and diameter (d_{LID}), respectively, of at least about 90 mm. Element 50: wherein the fracturing string has an outside diameter (d_{FOD}) of at least about 78 mm. Element 51: wherein the fracturing string has an outside diameter (d_{FOD}) of at least about 85.7 mm. Element 52: wherein the fracturing string has an inside diameter (d_{FID}) of at least about 50.8 mm. Element 53: wherein the multilateral junction is operable to handle at least 8,000 psi burst rate. Element 54: wherein the multilateral junction is operable to handle at least 10,000 psi burst rate. Element 55: wherein the second and third centerlines are angled relative to one another. Element 56: wherein the second and third bores overlap one another proximate the single first bore. Element 57: wherein a shared interior wall of the second and third bores comes to a sharp point at a location wherein the second and third bores overlap one another. Element 58: wherein a shared interior wall of the

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second and third bores comes to a blunt stress relief point at a location wherein the second and third bores overlap one another.

Those skilled in the art to which this application relates will appreciate that other and further additions, deletions, substitutions and modifications may be made to the described embodiments.

What is claimed is:

1. A y-block, comprising:
 - a housing comprising a monolithic block of material, the housing having a first end and a second opposing end defining a length (L);
 - a single first bore extending into the monolithic block of material from the first end, the single first bore defining a first centerline; and
 - second and third separate bores extending into the monolithic block of material and branching off from the single first bore, the second bore defining a second centerline and the third bore defining a third centerline, wherein the second and third centerlines are angled relative to one another, wherein the second and third bores include a shared interior wall along at least a portion of the length (L), the shared interior wall extending to a downhole end of at least one of the second or third separate bores, wherein the second bore includes a pin joint at the second opposing end and the third bore includes a box joint at the second opposing end.
2. The y-block as recited in claim 1, wherein the second centerline is angled relative to the first centerline.
3. The y-block as recited in claim 2, wherein the third centerline is angled relative to the first centerline.
4. The y-block as recited in claim 3, wherein the second centerline has a greater angle (θ) between itself and the first centerline than an angle (α) between the third centerline and the first centerline.
5. The y-block as recited in claim 4, wherein the second bore is a main leg bore and the third bore is a lateral leg bore.
6. The y-block as recited in claim 1, wherein the second bore has a diameter (d_2) and the third bore has a diameter (d_3), and further wherein the diameter (d_2) is the same as the diameter (d_3).
7. The y-block as recited in claim 6, wherein the second bore has a first portion having the diameter (d_2) and a second portion having a greater diameter (d_2).
8. The y-block as recited in claim 7, wherein the second portion is located between the first portion and the single first bore.
9. The y-block as recited in claim 8, wherein a length (L_2) of the second portion is at least two times a length (L_1) of the first portion.
10. The y-block as recited in claim 1, wherein the second and third bores overlap one another proximate the single first bore.
11. The y-block as recited in claim 10, wherein the shared interior wall of the second and third bores comes to a sharp point at a location wherein the second and third bores overlap one another.
12. The y-block as recited in claim 10, wherein the shared interior wall of the second and third bores comes to a blunt stress relief point at a location wherein the second and third bores overlap one another.
13. A multilateral junction, comprising:
 - a y-block, the y-block including;
 - a housing comprising a monolithic block of material, the housing having a first end and a second opposing end defining a length (L);

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a single first bore extending into the monolithic block of material from the first end, the single first bore defining a first centerline; and

second and third separate bores extending into the monolithic block of material and branching off from the single first bore, the second bore defining a second centerline and the third bore defining a third centerline, wherein the second and third centerlines are angled relative to one another, wherein the second and third bores include a shared interior wall along at least a portion of the length (L), the shared interior wall extending to a downhole end of at least one of the second or third separate bores, wherein the second bore includes a pin joint at the second opposing end and the third bore includes a box joint at the second opposing end;

a mainbore leg coupled to one of the second bore or the third bore for extending into a main wellbore; and
a lateral bore leg coupled to an other of the third bore or the second bore for extending into a lateral wellbore.

14. The multilateral junction as recited in claim 13, wherein the mainbore leg and the lateral bore leg are threadingly engaged with the y-block.

15. The multilateral junction as recited in claim 13, wherein the second bore and the third bore each include only a single straight centerline.

16. The multilateral junction as recited in claim 13, wherein the second centerline is angled relative to the first centerline.

17. The multilateral junction as recited in claim 16, wherein the third centerline is angled relative to the first centerline.

18. The multilateral junction as recited in claim 17, wherein the second centerline has a greater angle (θ) between itself and the first centerline than an angle (α) between the third centerline and the first centerline.

19. The multilateral junction as recited in claim 18, wherein the second bore is a main leg bore and the third bore is a lateral leg bore.

20. The multilateral junction as recited in claim 13, wherein the second bore has a diameter (d_2) and the third bore has a diameter (d_3), and further wherein the diameter (d_2) is the same as the diameter (d_3).

21. The multilateral junction as recited in claim 20, wherein the second bore has a first portion having the diameter (d_2) and a second portion having a greater diameter (d_2).

22. The multilateral junction as recited in claim 21, wherein the second portion is located between the first portion and the single first bore.

23. The multilateral junction as recited in claim 22, wherein a length (L_2) of the second portion is at least two times a length (L_1) of the first portion.

24. The multilateral junction as recited in claim 13, wherein the second and third bores overlap one another proximate the single first bore.

25. The multilateral junction as recited in claim 24, wherein the shared interior wall of the second and third bores comes to a sharp point at a location wherein the second and third bores overlap one another.

26. The multilateral junction as recited in claim 24, wherein the shared interior wall of the second and third bores comes to a blunt stress relief point at a location wherein the second and third bores overlap one another.

27. A well system, comprising:

- a main wellbore;
- a tubing string located in the main wellbore

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a lateral wellbore extending from the main wellbore; and
 a multilateral junction positioned at an intersection of the
 main wellbore and the lateral wellbore, the multilateral
 junction including;

a y-block located in the tubing string, the y-block
 including;

a housing comprising a monolithic block of material,
 the housing having a first end and a second
 opposing end defining a length (L);

a single first bore extending into the monolithic
 block of material from the first end, the single first
 bore defining a first centerline; and

second and third separate bores extending into the
 monolithic block of material and branching off
 from the single first bore, the second bore defining
 a second centerline and the third bore defining a
 third centerline, wherein the second and third
 centerlines are angled relative to one another,
 wherein the second and third bores include a
 shared interior wall along at least a portion of the
 length (L), the shared interior wall extending to a
 downhole end of at least one of the second or third
 separate bores, wherein the second bore includes
 a pin joint at the second opposing end and the third
 bore includes a box joint at the second opposing
 end;

a mainbore leg coupled to one of the second bore or the
 third bore and extending into the main wellbore; and
 a lateral bore leg coupled to an other of the third bore
 or the second bore and
 extending into the lateral wellbore.

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28. The well system as recited in claim 27, wherein the
 mainbore leg and the lateral bore leg are threadingly
 engaged with the y-block.

29. The well system as recited in claim 27, wherein the
 second bore and the third bore each include only a single
 straight centerline.

30. The well system as recited in claim 27, wherein the
 second centerline is angled relative to the first centerline and
 the third centerline is angled relative to the first centerline.

31. The well system as recited in claim 30, wherein the
 second centerline has a greater angle (θ) between itself and
 the first centerline than an angle (α) between the third
 centerline and the first centerline.

32. The well system as recited in claim 27, wherein the
 second bore has a first portion having a diameter (d_2) and a
 second portion having a greater diameter (d_2').

33. The well system as recited in claim 32, wherein the
 second portion is located between the first portion and the
 single first bore.

34. The well system as recited in claim 27, wherein the
 second and third bores overlap one another proximate the
 single first bore.

35. The well system as recited in claim 34, wherein the
 shared interior wall of the second and third bores comes to
 a sharp point at a location wherein the second and third bores
 overlap one another.

36. The well system as recited in claim 34, wherein the
 shared interior wall of the second and third bores comes to
 a blunt stress relief point at a location wherein the second
 and third bores overlap one another.

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